IOP Publishing
Phys. Med. Biol. 71 (2026) 125020
https://doi.org/10.1088/1361-6560/ae7bac

Physics in Medicine & Biology

Check for updates

RECEIVED
5 January 2026

REVISED
15 May 2026

ACCEPTED FOR PUBLICATION
10 June 2026

PUBLISHED
24 June 2026

PAPER

ROI-aware uncertainty fusion for label-efficient glioma MRI segmentation

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Keywords: active learning, glioma MRI segmentation, uncertainty estimation, borda count, roi-aware acquisition, rank-level fusion

Abstract

Accurate glioma MRI segmentation is critical for clinical decision-making, yet voxel-wise annotation of volumetric MR images is expensive and time-consuming. We aim to maximize segmentation quality while reducing annotation cost through selective labeling. We develop a tumor-focused active learning AL framework that mirrors real-world annotation workflows and combines three complementary techniques: (i) a patient-specific, tumor-focused 40-slice window to preserve context while reducing computational requirements; (ii) a *diverse* uncertainty committee spanning softmax entropy, margin, least confidence, variation ratio, and Bayesian AL by disagreement (BALD), where variation ratio and BALD are estimated via lightweight Monte Carlo dropout to capture epistemic uncertainty; and (iii) *multi-criteria rank-level fusion* via Borda count to robustly aggregate uncertainty signals. In particular, we compute uncertainty both at the *slice* level and at the *region-of-interest (ROI)* level by summarizing pixel-wise uncertainty within tumor ROIs, then fuse all techniques and granularities with Borda rank to form a single acquisition order. A segmentation network is trained on an initial small labeled subset and then iteratively selects additional slices for annotation. Evaluation on UCSF–PDGM and BraTS 2019 under realistic annotation budgets shows strong label efficiency and robustness: on UCSF–PDGM, near-full performance is achieved with only $\approx 16\%$ of the training data; on BraTS 2019, using $\approx 32\%$ of the data yields performance comparable to models trained on all data, even surpassing the full-data model. Overall, the proposed framework combines the tumor-focused window with MC-dropout-enhanced, *ROI-aware* uncertainty and Borda fusion to achieve accurate segmentation while aligning with real-world curation workflows.

1. Introduction

Volumetric glioma MRI annotation for tumor segmentation presents substantial clinical challenges, requiring expert radiologists to perform voxel-wise delineation across hundreds of slices per patient. This process is both time-intensive and costly, with a single BraTS case demanding 2–4 hours of expert annotation (Menze *et al* 2015). While modern deep learning architectures typically require large labeled samples; although data-efficient alternatives (e.g. self/semi-supervised learning or augmentation) help, the most direct lever is to change *what* we label via *active learning* (AL). In AL, a model is trained on a small seed set, then the unlabeled pool is scored with an acquisition function, and only the most informative samples are selected for annotation; this train→score→label loop iterates under a fixed budget (Settles 2009, Budd *et al* 2019).

However, applying AL to glioma MRI segmentation reveals several persistent research gaps. First, many medical AL methods suffer from *label leakage* or *confirmation bias*, where acquisition functions inadvertently rely on ground-truth information or model-generated pseudo-labels during selection (Boehringer *et al* 2023). Second, *instability and redundancy* plague single-metric uncertainty approaches, as different uncertainty measures (entropy, margin, Bayesian AL by disagreement BALD) often select divergent samples, while top-K sampling frequently acquires redundant, highly correlated slices (Sener and Savarese 2018, Kirsch *et al* 2019, Ash 2020, Kim *et al* 2024). Third, *background domination* occurs in class-imbalanced medical images, where non-tumor regions overwhelm uncertainty signals, causing the algorithm to select slices based on irrelevant background variations rather than pathological regions (Ma *et al* 2024). Fourth, existing methods often ignore the *clinical workflow constraints*, failing to account for the natural tumor distribution across slices or the radiologist's preference for reviewing contiguous, clinically relevant regions (Zhou *et al* 2024). Finally, most approaches lack *robust multi-criteria fusion*, leaving the practitioner to choose among competing uncertainty metrics without principled aggregation methods (Gaillochet *et al* 2023).

To address these limitations, we present a task-specific integration of established active-learning components into a label-free, tumor-focused AL framework for glioma MRI segmentation, combining three complementary elements: (1) a patient-specific 40-slice window centered on the slice of maximal tumor area to preserve clinical context while reducing computational load; (2) *ROI-aware* uncertainty that aggregates entropy, margin/least-confidence, and MC-dropout–based variation ratio/BALD *inside the predicted tumor* to mitigate background bias; and (3) rank-level fusion via Borda count to unify heterogeneous uncertainty signals into a stable acquisition decision. Starting from a small labeled subset, the model iteratively scores the unlabeled pool, fuses per-slice and per-ROI ranks, acquires the top-K slices for annotation without consulting ground truth, and repeats until the label budget is met.

Empirically, the proposed approach achieves near-full performance for glioma MRI segmentation with ∼16% labels on UCSF–PDGM and ∼32% on BraTS 2019, matching or slightly exceeding full-data training while improving stability across runs.

Building on strong encoder–decoder backbones such as U-Net (Ronneberger *et al* 2015) and validated against established glioma benchmarks (Menze *et al* 2015), our contributions highlight how a task-aware slice window, ROI-localized uncertainty, and rank-level fusion can deliver label-efficient, robust acquisition under realistic annotation constraints. Because both UCSF–PDGM and BraTS 2019 are glioma-focused datasets, the scope of this study is label-efficient *glioma MRI segmentation*. The contribution is the integration of tumor-focused windowing, ROI-aware uncertainty aggregation, and rank-level fusion into a practical active-learning workflow for this setting.

The rest of the paper is organized as follows: section 2 reviews related work in AL for medical image segmentation, with particular focus on brain tumor MRI applications, uncertainty estimation methods, batch selection strategies, and structure-aware acquisition techniques. Section 3 details our proposed tumor-focused AL framework, including the 40-slice windowing strategy, ROI-aware uncertainty computation, and Borda count rank fusion. Section 4 presents our experimental setup, datasets, and comprehensive evaluation results from all three experiments. Section 5 provides a detailed analysis and discussion of our findings. Finally, section 6 summarizes our contributions and outlines future research directions.

2. Related Work

2.1. Dataset-scale AL for brain tumor MRI

AL for brain MRI segmentation has been explored at the cohort scale to reduce annotation cost while maintaining diagnostic accuracy. Boehringer *et al* trained HighRes3DNet on BraTS-2021 (1,251 cases) and implemented a *performance-gated* loop that ranked unlabeled scans by predicted Dice (2023). Cases with Dice $\geqslant 0.7$ were pseudo-labeled, while low-confidence cases were sent to experts. This achieved Dice 0.868 with only 28.6% manual labels compared with 0.906 under full supervision, cutting annotation by more than two-thirds. Despite its simplicity, the approach highlighted two practical risks relevant to our work: confirmation bias from reusing model-generated masks and domain drift when pseudo-labels dominate.

Kim *et al* conducted a 638-patient benchmark with a 3D U-Net trained across fifteen random seeds (2024). They compared random, margin, ensemble, and MC-dropout strategies, introducing a redundancy-restriction filter that enforces feature-space dissimilarity between acquisitions. MC-dropout with redundancy control achieved full-data performance with only 17.5%–30% labeled data, whereas naive margin sampling occasionally underperformed random selection. Their analysis underscored the influence of initialization variance and the importance of repeated trials for reliable AL evaluation, directly motivating our use of multi-criteria fusion to improve stability.

2.2. Uncertainty: epistemic vs aleatoric; BALD, BatchBALD, and clinical interpretation

A key design question in AL is which form of uncertainty to model. MonteCarlo dropout offers an efficient estimate of *epistemic* uncertainty without ensembles, enabling mutual-information metrics such as BALD (Gal and Ghahramani 2016, Gal *et al* 2017). Epistemic uncertainty dominates early in training when data are scarce, while predictive entropy becomes smoother yet less selective at later stages. BatchBALD extends the BALD criterion to batch selection by jointly estimating mutual information across samples, reducing redundancy in co-acquisitions (Kirsch *et al* 2019). More broadly, uncertainty in medical image analysis should not be treated as a single phenomenon tied only to model confidence. A recent review by Huang *et al* emphasizes that uncertainty can arise from multiple sources, including limited training data, imperfect model specification, acquisition noise, annotation ambiguity, and post-processing assumptions, and that these sources may interact in practice rather than appear in isolation (2024). This broader view is important in medical AL, because an acquisition score may reflect not only lack of data but also ambiguity in the image formation and annotation process itself.

From this broader perspective, aleatoric uncertainty refers to ambiguity inherent to the image and label generation process, whereas epistemic uncertainty reflects uncertainty in the model parameters due to limited training data or insufficient coverage of the data distribution. In segmentation settings, both may appear simultaneously in the same region, especially near lesion boundaries or in low-quality images. This distinction matters because uncertainty-driven acquisition rules are often motivated by epistemic uncertainty, while clinicians may visually interpret the resulting maps as if they directly reflect anatomical ambiguity (Huang *et al* 2024). A concrete example of this distinction was provided by Wang *et al*, who estimated aleatoric uncertainty through test-time augmentation for medical image segmentation (2019). Their results showed that prediction variability can arise not only from model uncertainty but also from sensitivity to plausible input perturbations, thereby linking uncertainty to image quality, acquisition variability, and local ambiguity in the data itself. For our study, this literature is important because it cautions against interpreting every uncertain region as purely a consequence of insufficient labels or incomplete training. A second clinically relevant source of uncertainty is disagreement introduced by the annotation and image-processing pipeline. Jungo *et al* showed that inter-observer variability can substantially affect uncertainty estimation in medical image segmentation, meaning that uncertain predictions may partly reflect disagreement among experts rather than only weakness of the model (2018). Likewise, Bierbrier *et al* reviewed uncertainty arising from medical image registration and confidence estimation, highlighting that preprocessing, alignment errors, and transformation assumptions can propagate into downstream quantitative analyses (2022). These findings are directly relevant to MRI tumor segmentation, where manual boundaries are often difficult to define sharply and multimodal alignment quality influences voxel-level interpretation. Brain tumor segmentation provides a particularly important setting for such caution. In the QU-BraTS uncertainty challenge analysis, Mehta *et al* demonstrated that uncertainty estimation in brain tumor segmentation is strongly affected by the complexity of tumor subregions and by differences in model calibration and ranking behavior across methods (2022). More generally, Faghani *et al* argued from a radiology perspective that uncertainty information becomes clinically meaningful only when it is tied to interpretable causes and decision support, rather than reported as a generic confidence map (2023). Accordingly, in our work we view uncertainty maps primarily as *annotation-prioritization* and *reliability* signals within the active-learning loop, not as stand-alone clinical biomarkers. This framing better aligns the uncertainty literature with the practical goal of reducing annotation burden in glioma MRI segmentation.

2.3. Batch selection and redundancy control

A common pitfall in AL is the tendency of top-K uncertainty sampling to select highly correlated and redundant samples. Various strategies have been developed to address this challenge. Geometry-based methods like core-set coverage optimize representativeness in feature space (Sener and Savarese 2018), while gradient-based approaches such as BADGE promote diversity through gradient embeddings (Ash 2020). Alternative strategies include adversarial methods like VAAL (variational adversarial active learning) and surrogate-driven approaches like Learning Loss, which learn sample informativeness indirectly but often introduce additional model complexity (Sinha *et al* 2019, Yoo and Kweon 2019).

Gaillochet *et al* introduced the *Stochastic Batches (SB)* strategy (2023), which repeatedly forms random candidate batches, scores each by mean uncertainty (entropy, MC-dropout, test-time augmentation, or learning-loss), and selects the batch with the highest score. On PROMISE12 and Hippocampus datasets, SB improved Dice and Hausdorff metrics while lowering redundancy across multiple random seeds. This highlights the benefit of stochastic diversification and score averaging as lightweight alternatives to explicit diversity models.

2.4. Structure-/ROI-aware selection and minimal-label protocols

Spatial structure plays a crucial role in medical segmentation, motivating AL methods that embed shape and boundary priors into the acquisition rule. Zhou *et al* introduced **SBC-AL**, which selects images with low structural and boundary consistency between predicted and refined masks (2024). On ACDC and KiTS19, SBC-AL exceeded entropy-, margin-, and VAAL-based baselines, achieving over 95% of full-data Dice with only 16.7% labeled data. Ma *et al* proposed **SU-AL**, which filters uncertainty maps to emphasize lesion and boundary pixels prior to aggregation (2024). On BraTS and MSD-Spleen, SU-AL reached full-supervision Dice with about 13% labeled slices, halving annotation demand relative to plain uncertainty sampling. Wu *et al* extended these ideas to 3D through **MF-ConS**, combining teacher–student consistency and active cross-annotation (2024). Each iteration selected low-consistency scans and annotated only three orthogonal slices per volume, achieving 81%–84% Dice at 10%–20% label budgets while annotating under 0.5% of voxels. Collectively, these studies demonstrate that incorporating structure-awareness, slice sparsity, and redundancy control enables high segmentation fidelity from limited annotations. Taken together, these studies suggest that uncertainty is most useful when it is localized, structure-aware, and interpreted in the context of the imaging and annotation pipeline rather than treated as a purely abstract model score. This observation directly motivates our ROI-level aggregation and rank-fusion design. In the present study, these recent methods are used primarily to position and motivate the proposed framework, whereas the direct experimental comparison focuses on representative uncertainty-based acquisition criteria and their fusion-based combinations under a common protocol.

3. Proposed AL framework

3.1. Overview and architecture

We present an end-to-end AL framework that addresses the annotation bottleneck in glioma MRI segmentation by iteratively selecting the most informative slices for expert labeling. As illustrated in figure 1, multi-modal MR volumes are preprocessed by removing empty slices and applying per-channel z-score normalization. For each patient, we form a 40-slice, **tumor-focused** window centered on the slice with the maximal tumor area. Each slice (4 modalities $\times H \times W$) is fed to a 2D U-Net-style encoder-decoder (Ronneberger *et al* 2015) comprising encoder blocks of 3×3 Conv-BN-ReLU with stride-2 downsampling, skip connections into a decoder with bilinear upsampling and 3×3 refinement convolutions, and a final 1×1 convolutional head with sigmoid/softmax to produce a per-pixel tumor probability map $p_x \in [0,1]$ (binary) or $p_x \in \Delta^{C-1}$ (multi-class). In the experiments reported in this paper, we instantiate the framework in its binary tumor-versus-background form to maintain a consistent target across datasets and to focus the evaluation on annotation-efficient acquisition rather than subregion-specific prediction. The core AL loop begins by training an initial model on a small, stratified labeled subset. This model then processes the unlabeled data to compute pixel-wise uncertainties using a diverse committee of uncertainty quantification methods. These uncertainties are aggregated at both slice and region-of-interest (ROI) levels, then fused via Borda count to create a unified ranking. The top-K most uncertain slices are selected for expert annotation, added to the training set, and the model is retrained. This process iterates until the annotation budget is exhausted. Although the individual building blocks of the framework are established in the active-learning literature, their combination in our setting is deliberate and task-driven. The 40-slice tumor-focused window addresses workflow efficiency and anatomical relevance, ROI-aware aggregation reduces background-dominated uncertainty in highly imbalanced slices, and Borda rank fusion stabilizes acquisition by combining heterogeneous uncertainty criteria without assuming a shared numeric scale. In this sense, the contribution is best understood as an integrated and practically oriented acquisition framework rather than as a stand-alone new uncertainty formulation.

3.2. Preprocessing and tumor-focused window

All MRI scans are preprocessed to ensure consistency across patients and imaging modalities. Each sequence is aligned to a common reference space and intensity-normalized to reduce scanner-related variations. Non-informative empty slices are discarded to avoid unnecessary computation, and the

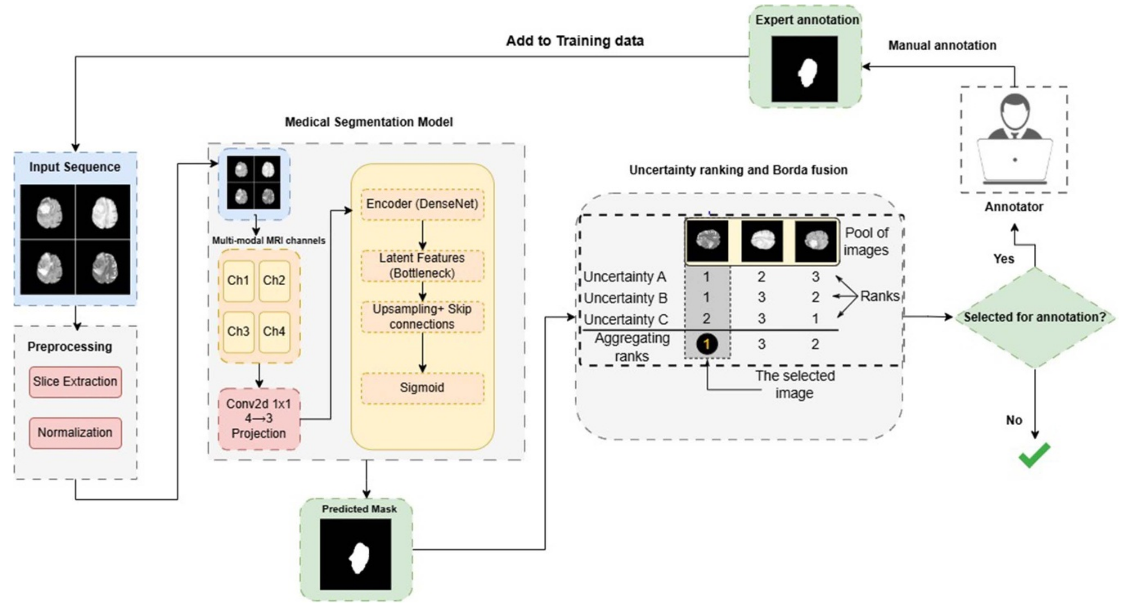


Figure 1. Overview of the proposed active learning framework for glioma MRI segmentation.

remaining slices are standardized using per-channel *z*-score normalization to achieve comparable intensity distributions across modalities.

To balance efficiency and anatomical completeness, a *tumor-focused window* is defined for each patient. The slice with the largest tumor area is first identified from the segmentation mask, and a fixed range of 40 axial slices (20 above and 20 below) is extracted around it. This window captures the entire tumor region along with its surrounding healthy tissue, providing sufficient spatial context for boundary learning while removing irrelevant cranial and caudal sections. All subsequent training, inference, and active-learning selection steps are performed within this 40-slice window. This strategy directly addresses the clinical workflow constraint by focusing annotation efforts on clinically relevant regions.

3.3. Segmentation backbones

To identify an optimal feature extractor for our segmentation framework, we evaluated five widely used CNN backbones–ResNet50 (He *et al* 2016), MobileNetV2 (Sandler *et al* 2018), DenseNet121 (Huang *et al* 2017), VGG16 (Simonyan and Zisserman 2014), and EfficientNetB0 (Tan and Le 2019)–each coupled with an identical U-Net style decoder. All models were trained under the same optimization settings, epoch schedules, and slice budgets to ensure a fair comparison.

This systematic backbone screening allowed us to analyze trade-offs between representational strength and computational efficiency in the context of limited labeled data. ResNet50 and VGG16 provided strong spatial localization, while MobileNetV2 and EfficientNetB0 offered lightweight alternatives with lower inference cost. DenseNet121 was evaluated for its feature reuse properties and parameter efficiency. The comparative analysis (detailed in section 4) assessed each backbone's performance across both 40-slice subsets and full-volume experiments under constrained data conditions to inform our final encoder selection.

Training objective: For binary tumor-versus-background segmentation, the network is trained using Dice loss, which directly optimizes overlap between the predicted mask and the ground-truth mask in the class-imbalanced foreground/background setting. In the implementation used for the reported experiments, the model is optimized with Adam using a learning rate of 10^{-4},

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_i y_i \hat{y}_i + \epsilon}{\sum_i y_i + \sum_i \hat{y}_i + \epsilon} \tag{1}$$

where y_i and $\hat{y}_i$ denote the ground-truth and predicted mask values at pixel i, respectively, and $\epsilon = 10^{-6}$ is a smoothing constant.

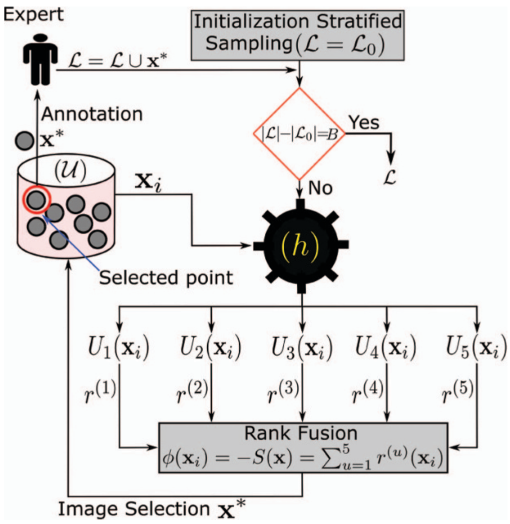


Figure 2. Active learning cycle for glioma MRI segmentation. The process begins with a small initial labeled set $\mathcal{L}_0$ obtained through stratified sampling. In each iteration: (1) a segmentation model is trained on the current labeled set $\mathcal{L}$, (2) uncertainty is computed on the unlabeled pool $\mathcal{U}$ using our multi-criteria committee, (3) the most informative slice $\mathbf{x}^*$ is selected via Borda count fusion, and (4) the expert-annotated slice is added to $\mathcal{L}$. This loop continues until the annotation budget is exhausted, maximizing performance under limited labeling resources.

3.4. Active learning framework and uncertainty quantification

Our work is situated within the context of pool-based AL, a paradigm designed to maximize model performance under a limited annotation budget. We begin with a large pool of unlabeled volumetric MRI data, denoted as $\mathcal{U} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{n_u}\}$, where each $\mathbf{x}_i$ is an unlabeled scan and n_u is the total number of unlabeled volumes. The labeled set $\mathcal{L}$ is initially empty ($\mathcal{L} = \emptyset$). A fixed annotation budget B constrains the total number of data points (in our case, 2D slices) that can be selected for labeling by an expert (see figure 2).

AL is an iterative process that combines this small labeled set $\mathcal{L}$ with the large unlabeled pool $\mathcal{U}$. In each cycle, a model is trained on $\mathcal{L}$ and then used to evaluate all instances in $\mathcal{U}$. The core of an AL strategy is an acquisition function $\phi(\cdot)$, which quantifies the informativeness of an unlabeled instance $\mathbf{x}$ relative to the current model h. The most informative sample is selected as:

$$\mathbf{x}^* = \arg\max_{\mathbf{x} \in \mathcal{U}} \phi(\mathbf{x}, h). \tag{2}$$

This instance $\mathbf{x}^*$ is then labeled and added to $\mathcal{L}$ ($\mathcal{L} = \mathcal{L} \cup \mathbf{x}^*$), removed from $\mathcal{U}$ ($\mathcal{U} = \mathcal{U} \setminus \mathbf{x}^*$), and the model is retrained. This loop continues until the budget B is exhausted.

The effectiveness of this process hinges entirely on the design of $\phi(\cdot)$. To overcome the limitations of single-metric approaches, we frame the acquisition function around uncertainty quantification, aiming to identify the slices where the current segmentation model is least confident. To this end, our framework employs a diverse committee of uncertainty estimation techniques. We evaluate both *Traditional* approaches, which rely on single deterministic forward passes, and *Advanced* approaches that leverage Monte Carlo (MC) dropout to capture model (epistemic) uncertainty (Gal and Ghahramani 2016, Gal *et al* 2017). While the following methods are described for binary tumor segmentation, they extend naturally to multi-class settings. We use the binary formulation in this study for methodological focus. The acquisition mechanism proposed here is designed to rank slices according to tumor presence, tumor-centered uncertainty, and annotation utility under a restricted expert budget. For this purpose, a whole-tumor foreground is sufficient to define the 40-slice tumor-focused window, compute ROI-level uncertainty summaries, and compare acquisition strategies consistently across datasets. This choice reduces confounding factors introduced by dataset-specific subregion definitions and enables a cleaner evaluation of the active-learning framework itself. At the same time, the formulation does not preclude multi-class use; the same acquisition logic can be extended to subregion-aware settings by computing class-specific probabilities and uncertainty summaries for multiple foreground classes.

3.4.1. Initial selection and stratified sampling

Before the AL cycle can begin, the initial labeled set $\mathcal{L}_0$ must be populated. Since the choice of this initial set can significantly impact the subsequent acquisition trajectory, we employ a structured approach to ensure a fair and reproducible comparison across all strategies.

To this end, we establish an identical initial training set through stratified sampling based on tumor burden. For each patient, the tumor area is computed from the ground-truth mask. Based on the proportion of tumor pixels within the brain region, each slice is assigned to one of four categories: **no tumor** ($A_{\text{tumour}} = 0$), **small** ($0 < A_{\text{tumour}} \leqslant 25\%$), **medium** ($25\% < A_{\text{tumour}} \leqslant 50\%$), and **large** ($A_{\text{tumour}} > 50\%$). A fixed random seed is then used to sample a balanced number of slices from each category, producing an initial labeled subset $\mathcal{L}_0$ that maintains diversity in tumor size and location. This initial set serves as the starting point for the AL loop, ensuring that all methods begin from an equivalent state.

The selected indices are stored in a dedicated CSV file and reused across all experiments to guarantee consistent initialization. Two random baselines are reported for fairness: (i) a *controlled random* baseline that uses the same stratified initial set as all active-learning strategies, and (ii) a *pure random* baseline that independently samples a new random subset for each run. This separation isolates the influence of the acquisition policy from that of the starting distribution.

3.4.2. Uncertainty estimation techniques

Given the initialized model trained on $\mathcal{L}_0$, the core of our AL strategy lies in the acquisition function $\phi(\cdot)$, which we design around a diverse set of uncertainty estimation techniques. We leverage both traditional single-pass methods and advanced approaches using MC dropout to capture model (epistemic) uncertainty.

Traditional methods (single forward pass): For a model h performing binary tumor segmentation, let $p_t(\mathbf{x}_i)$ be the predicted probability of tumor at pixel i in an unlabeled slice $\mathbf{x}_i$, and $p_b(\mathbf{x}_i) = 1 - p_t(\mathbf{x}_i)$ the background probability. We compute three established pixel-wise uncertainty measures:

- **Entropy:** $H(\mathbf{x}) = -\sum_{c\in t,b} p_c(\mathbf{x}) \log p_c(\mathbf{x})$,
- **Least confidence:** $LC(\mathbf{x}) = 1 - \max(p_t(\mathbf{x}), p_b(\mathbf{x}))$,
- **Margin:** $M(\mathbf{x}) = p_1(\mathbf{x}) - p_2(\mathbf{x})$, where p_1 and p_2 are the highest and second-highest class probabilities respectively.

These measures capture different aspects of model uncertainty: entropy measures the overall unpredictability, least confidence focuses on the maximum probability, and margin captures the difference between the top two class probabilities. For AL acquisition, we aggregate these pixel-wise scores to the slice level by computing the average uncertainty across all pixels in a slice. We then seek slices with the highest average entropy, highest average least confidence, and lowest average margin (since a smaller margin indicates higher uncertainty).

Advanced methods (MC dropout): While traditional methods quantify overall prediction uncertainty, they cannot distinguish between inherent data ambiguity and the model's lack of knowledge. To specifically target the latter–known as *epistemic uncertainty*–we employ MC dropout during inference. By performing T stochastic forward passes with dropout active, we obtain multiple probability estimates $\{p_{\mathbf{x}i,t}\}_{t=1}^{T}$ for each pixel $\mathbf{x}_i$, effectively sampling from the model's posterior distribution. This approach allows us to measure how much the model's predictions vary due to parameter uncertainty, which directly indicates where additional training data would be most beneficial. We compute four advanced uncertainty measures:

- **MC entropy (MC average):** $H_{\text{MC}}(\mathbf{x}) = \frac{1}{T}\sum_{t=1}^{T} h\big(p_{\mathbf{x},\mathbf{t}}\big)$.
- **MC margin:** $M_{\text{MC}}(\mathbf{x}) = \frac{1}{T}\sum_{t=1}^{T}\Big(1 - |2p_{\mathbf{x},\mathbf{t}} - 1|\Big) = \frac{2}{T}\sum_{t=1}^{T}\min\{p_{\mathbf{x},t}, 1 - p_{\mathbf{x},t}\}$,
- **Variation ratio:** $VR(\mathbf{x}) = 1 - \frac{1}{T}\max\{N_1(\mathbf{x}), N_0(\mathbf{x})\}$, where $N_1(\mathbf{x}) = \sum_{t=1}^{T}\mathbf{1}\{p_{\mathbf{x},t} \geqslant 0.5\}$ and $N_0(\mathbf{x}) = T - N_1(\mathbf{x})$. This measures the **lack of consensus** among forward passes, with higher values indicating greater disagreement.
- **BALD:** $\text{BALD}(\mathbf{x}) = h(\hat{p}_{\mathbf{x}}) - \frac{1}{T}\sum_{t=1}^{T} h\big(p_{\mathbf{x},\mathbf{t}}\big)$, where $\hat{p}_{\mathbf{x}} = \frac{1}{T}\sum_{t=1}^{T} p_{\mathbf{x},\mathbf{t}}$ and $h(p) = -p\log_2 p - (1-p)\log_2(1-p)$ is the binary entropy function. BALD is capturing the mutual information between model parameters and predictions (Gal and Ghahramani 2016, Gal et al 2017).

3.4.3. Illustrative example

To clarify the difference between traditional and MC-Dropout uncertainty estimation, consider a binary tumor segmentation task analyzing one specific pixel. Given:

$$p_t = 0.85, \quad p_b = 0.15.$$

Table 1. MC Dropout forward passes with entropy calculations for each pass.

Pass (t)	p_t	p_b	Prediction	Entropy (H_t)
1	0.82	0.18	Tumor ($p_t > 0.5$)	0.66
2	0.15	0.85	Background ($p_t < 0.5$)	0.61
3	0.90	0.10	Tumor	0.47
4	0.88	0.12	Tumor	0.50
5	0.10	0.90	Background	0.47

Traditional method (single forward pass, dropout disabled during inference) provides a single probability estimate. The uncertainty calculations are:

$$\begin{aligned} H &= -[p_t \log_2 p_t + p_b \log_2 p_b] = -[0.85 \times \log_2(0.85) + 0.15 \times \log_2(0.15)] = 0.61 \\ \mathrm{LC} &= 1 - \max(p_t, p_b) = 1 - 0.85 = 0.15 \\ M &= |p_t - p_b| = |0.85 - 0.15| = 0.70 \\ &\Rightarrow 1 - M = 1 - 0.70 = 0.30 \end{aligned}$$

Note that the margin is inverted $(1 - M)$ so that higher values consistently indicate higher uncertainty, making it compatible with Borda rank fusion.

MC-dropout method performs five stochastic forward passes ($T = 5$) with dropout enabled during inference, producing the following results in table 1.

The MC uncertainty measures are calculated as follows:

$$\text{MC entropy: } H_{\mathrm{MC}} = \frac{1}{5}\sum_{t=1}^{5} H_t = \frac{0.66 + 0.61 + 0.47 + 0.50 + 0.47}{5} = 0.54.$$

To calculate the VR uncertainty score, we first apply a decision threshold of 0.5 to each forward pass. For each pass where $p_t > 0.5$, the prediction is classified as Tumor; otherwise, it is classified as Background. From table 1, the passes 1, 3, and 4 are tumor predictions and the passes 2 and 5 are background predictions. The Variation Ratio is then calculated as:

$$\mathrm{VR} = 1 - \frac{\max(3, 2)}{5} = 1 - \frac{3}{5} = 0.40.$$

For BALD, we compute the difference between the entropy of the mean predictive probability and the mean entropy across the stochastic forward passes. The mean predictive probability is computed from the MC tumor probabilities, while the mean entropy is computed separately from the entropy values:

$$\begin{aligned} \hat{p}_t &= \frac{0.82 + 0.15 + 0.90 + 0.88 + 0.10}{5} = 0.57, \\ \hat{p}_b &= 1 - \hat{p}_t = 0.43, \\ H(\hat{p}) &= -[0.57 \times \log_2(0.57) + 0.43 \times \log_2(0.43)] = 0.98, \\ \frac{1}{5}\sum_{t=1}^{5} H_t &= \frac{0.66 + 0.61 + 0.47 + 0.50 + 0.47}{5} = 0.54, \\ \mathrm{BALD} &= H(\hat{p}) - \frac{1}{5}\sum_{t=1}^{5} H_t = 0.98 - 0.54 = 0.44. \end{aligned}$$

With multiple points, consider three pixels: $\mathbf{x}_1 : (0.85, 0.15)$, $\mathbf{x}_2 : (0.95, 0.05)$, $\mathbf{x}_3 : (0.80, 0.20)$. The uncertainty calculations are as follows, $H(\mathbf{x}_1) = 0.61, H(\mathbf{x}_2) = 0.25, H(\mathbf{x}_3) = 0.45$. The MC-dropout (5 forward passes) is as follows:

$$\begin{aligned} \mathbf{x}_1 &: \quad H_{\mathrm{MC}} = 0.54, \quad \mathrm{VR} = 0.40, \quad \mathrm{BALD} = 0.44 \\ \mathbf{x}_2 &: \quad H_{\mathrm{MC}} = 0.68, \quad \mathrm{VR} = 0.60, \quad \mathrm{BALD} = 0.52 \\ \mathbf{x}_3 &: \quad H_{\mathrm{MC}} = 0.42, \quad \mathrm{VR} = 0.20, \quad \mathrm{BALD} = 0.35 \end{aligned}$$

Table 2. Uncertainty scores and acquisition order for three pixels under different methods.

Method	x_1	x_2	x_3	Selection order
Traditional entropy	0.61	0.25	0.45	$\mathbf{x}_1 \rightarrow \mathbf{x}_3 \rightarrow \mathbf{x}_2$
MC entropy	0.54	0.68	0.42	$\mathbf{x}_2 \rightarrow \mathbf{x}_1 \rightarrow \mathbf{x}_3$
Variation ratio	0.40	0.60	0.20	$\mathbf{x}_2 \rightarrow \mathbf{x}_1 \rightarrow \mathbf{x}_3$
BALD	0.44	0.52	0.35	$\mathbf{x}_2 \rightarrow \mathbf{x}_1 \rightarrow \mathbf{x}_3$

Table 3. Named acquisition variants. MC dropout is used for the criteria that require it (VR/BALD).

Variant	Criteria fused by Borda	MC dropout
`EML`	Entropy + Margin + Least confidence (pixel-wise)	Off
`EMV`	Entropy (MC average) + Margin (MC average) + Variation ratio	On
`EVB`	Entropy (MC average) + Variation ratio + BALD	On

From the results in table 2, traditional entropy selects $\mathbf{x}_1$ as most uncertain, whereas the MC-dropout-based criteria prioritize $\mathbf{x}_2$ in this example. However, the relative ordering of the remaining pixels differs across criteria. This illustrates that different uncertainty measures can agree on the most informative point while still producing different acquisition rankings. This motivates the use of multi-criteria rank fusion rather than relying on any single uncertainty measure.

3.4.4. Multi-granularity aggregation and rank fusion

To bridge pixel-level uncertainties with slice-level acquisition decisions for the unlabeled pool $\mathcal{U}$, we aggregate uncertainty signals at two distinct granularities. For any pixel-wise uncertainty measure $U(\mathbf{x}_i)$ and a slice $\mathbf{x}$ with region of interest (ROI) $R_\mathbf{x}$ (the predicted tumor region), we compute:

$$U_{\text{slice}}(\mathbf{x}) = \frac{1}{|R_\mathbf{x}|}\sum_{\mathbf{x}_i \in R_\mathbf{x}} U(\mathbf{x}_i) \quad \text{(ROI-aware)}$$

$$U\text{full}(\mathbf{x}) = \frac{1}{|\mathbf{x}|}\sum_{\mathbf{x}_i \in \mathbf{x}} U(\mathbf{x}_i) \quad \text{(Whole-slice).}$$

If $R_\mathbf{x}$ is empty (no predicted tumor), $U_{\text{slice}}(\mathbf{x})$ defaults to $U_{\text{full}}(\mathbf{x})$. This dual approach emphasizes tumor-relevant regions while maintaining robustness.

To integrate selected uncertainty signals, we employ *multi-criteria rank-level fusion* at a chosen granularity level. For each selected uncertainty criterion u from our committee, we compute uncertainty using the slice granularity (ROI-aware aggregation) and rank all slices in the unlabeled pool $\mathcal{U}$ in descending order of uncertainty (except for margin-based measures, where ascending order is used). Each slice $\mathbf{x}$ receives a rank $r^{(u)}(\mathbf{x}) \in \{1, \ldots, |\mathcal{U}|\}$ for each criterion (see figure 2).

These ranks are aggregated via Borda count:

$$S(\mathbf{x}) = \sum_u r^{(u)}(\mathbf{x}). \tag{3}$$

The acquisition function $\phi(\mathbf{x})$ is then defined as the negative aggregate rank:

$$\phi(\mathbf{x}) = -S(\mathbf{x}) \tag{4}$$

and we select the slice that maximizes this function:

$$\mathbf{x}^* = \arg\max_{\mathbf{x}\in\mathcal{U}} \phi(\mathbf{x}) = \arg\max_{\mathbf{x}\in\mathcal{U}} [-S(\mathbf{x})]. \tag{5}$$

This rank fusion approach stabilizes acquisition decisions by leveraging complementary information across methods and granularities, making the selection process more robust than any single criterion alone. The Borda count mechanism effectively democratizes the diverse perspectives of our uncertainty committee.

Based on these components, we define three acquisition variants (table 3) that combine complementary uncertainty signals: EML (traditional), EMV (MC-enhanced with variation ratio), and EVB (MC-enhanced with mutual information).

Table 4. Rank assignments for three slices across four uncertainty criteria.

Slice	H	M	Variation ratio	BALD
$\mathbf{x}_1$	1	1	3	3
$\mathbf{x}_2$	3	3	1	2
$\mathbf{x}_3$	2	2	2	1

3.4.5. Borda count rank fusion example

To demonstrate how multiple uncertainty criteria are combined, consider three unlabeled slices $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ evaluated using four uncertainty measures: Entropy, Margin, Variation Ratio, and BALD. Each slice receives ranks from 1 (most uncertain) to 3 (least uncertain) for each criterion:

The Borda count aggregates these ranks from table 4 into a single score for each slice:

$$S(\mathbf{x}_1) = 1 + 1 + 3 + 3 = 8$$
$$S(\mathbf{x}_2) = 3 + 3 + 1 + 2 = 9$$
$$S(\mathbf{x}_3) = 2 + 2 + 2 + 1 = 7.$$

Since lower Borda scores indicate higher overall uncertainty, the acquisition order is:

$$\mathbf{x}^* = \arg\max_{\mathbf{x} \in \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}} - S(\mathbf{x}) = \mathbf{x}_3.$$

This demonstrates how Borda count identifies $\mathbf{x}_3$ as the most informative slice–not by being the most uncertain in any single criterion (it ranks first only in BALD), but by being **consistently uncertain across all criteria**. The method robustly combines diverse uncertainty signals to prioritize slices that will most improve model performance when labeled.

4. Experimental results

We conduct three main experiments to validate the efficacy and robustness of our proposed AL framework. The active-learning experiments are designed as controlled comparisons between representative uncertainty-based acquisition criteria and the proposed fusion-based acquisition variants, rather than as isolated internal ablations. Specifically, we evaluate established acquisition criteria from the active-learning literature—including entropy, least confidence, margin, variation ratio, and BALD—individually under the same initialization, backbone, annotation budget, and retraining protocol. We then compare these constituent methods against the proposed fusion-based variants (`EML`, `EMV`, and `EVB`) to assess whether combining complementary uncertainty signals through ROI-aware Borda-rank fusion improves label efficiency and acquisition robustness. In this way, the experiments examine both the individual behavior of widely used uncertainty criteria and the benefit of their principled integration within a unified evaluation framework. To assess sensitivity and stability, all active-learning strategies are additionally evaluated over repeated independent runs, and the results are analyzed using mean performance summaries, average-rank comparisons, pairwise consistency analysis, and ECDF-based views of variability under limited annotation budgets. The first experiment focuses on backbone selection and validation of the 40-slice protocol, where we compare five CNN architectures trained on 40-slice windows versus full-volume data to identify the optimal encoder for subsequent AL. Second, we assess AL strategies under a tight annotation budget, evaluating performance and stability on the UCSF-PDGM dataset across 11 repeated runs using only 16% of the available labels. Finally, we validate the framework's generalization on the BraTS 2019 dataset, testing multiple budget levels (approximately 32% and 16% of labels) and comparing the results against full-data performance.

For both UCSF–PDGM and BraTS 2019, we formulated the segmentation task as *binary tumor-versus-background* prediction by merging the available glioma subregions into a single foreground class. In BraTS, this means that enhancing tumor, necrotic/non-enhancing tumor core, and edema are treated jointly as one tumor label; in UCSF–PDGM, the available tumor annotations are likewise used to define a single foreground region. We adopted this formulation intentionally for the present study because our primary goal is to evaluate *label-efficient active-learning acquisition* under tight annotation budgets, rather than to optimize subregion-specific clinical delineation. A unified binary target also provides a consistent label space across the two datasets, which differ not only in imaging modalities but also in annotation structure and clinical emphasis. To reduce modality-related confounding in the cross-dataset comparison,

Table 5. Performance comparison between 40-slice subset and full-volume data across backbone models. Best values per column section are bolded.

	40-Slice Subset					Full Volume				
Model	Loss	IoU	F1	Precision	Recall	Loss	IoU	F1	Precision	Recall
ResNet50	0.1208	0.8230	0.8899	**0.9124**	0.8843	0.5820	0.3851	0.4278	0.4304	0.4465
MobileNet	0.1345	0.8019	0.8780	0.8644	0.9069	0.5800	0.3843	0.4290	0.4257	**0.4529**
DenseNet	**0.1191**	**0.8300**	**0.8981**	0.8845	**0.9222**	**0.5753**	0.3906	0.4318	0.4305	0.4499
VGG16	0.1203	0.8290	0.8926	0.8941	0.9032	0.5762	**0.3928**	**0.4320**	**0.4488**	0.4366
EffNetB0	0.1201	0.8263	0.8923	0.9062	0.8924	0.6089	0.3551	0.3998	0.4457	0.3855

we standardized both datasets to the same four selected MRI modalities and used them consistently as the four input channels of the network. Accordingly, the comparison is based on a matched four-channel input design rather than arbitrary modality mixtures. Nevertheless, the two datasets remain heterogeneous in acquisition protocol, scanner properties, patient population, and annotation context. Therefore, the cross-dataset results should be interpreted as evidence of robustness under matched-modality but heterogeneous imaging conditions, rather than as proof of full dataset equivalence. In our experiments, we employ four key metrics to assess segmentation quality:

- **Intersection over union (IoU)**: Measures the overlap between predicted and ground truth regions: $\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}}$, where TP, FP, FN denote true/false positives and false negatives. Higher values indicate better spatial alignment (Müller *et al* 2022).
- ***F*1-score (dice coefficient)**: The harmonic mean of precision and recall: $F1 = \frac{2\times\mathrm{TP}}{2\times\mathrm{TP}+\mathrm{FP}+\mathrm{FN}}$, providing a balanced measure of segmentation accuracy (Setiawan 2020).
- **Precision**: $\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}$, which quantifies the proportion of correct tumor predictions among all positive predictions (Müller *et al* 2022).
- **Recall**: $\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}$, measuring the proportion of actual tumor pixels correctly identified (Setiawan 2020, Müller *et al* 2022).

We evaluate at two granularities: **slice-level** metrics, computed by pooling all slice predictions, and **patient-level** metrics obtained by aggregating predictions across all slices per subject. The patient-level evaluation ensures clinical relevance.
Note on evaluation granularity: Throughout the paper, *Pool (slice-level) test* means evaluation over *all test slices pooled across subjects* (each slice is a sample). *Patient (per-subject) test* means evaluation on *full patients*, where predictions are aggregated across all slices of a subject and metrics are computed per subject (each patient counts once).

4.1. Backbone selection and 40-slice protocol validation

We first calibrated the segmentation backbone by comparing five CNN architectures on both tumor-focused 40-slice windows and full-volume data. This screening serves two purposes: identifying the optimal encoder for AL experiments and validating that our 40-slice protocol maintains accuracy while significantly reducing computational requirements.

As shown in table 5, DenseNet121 achieves the best performance on the 40-slice protocol with the lowest loss (0.1191), highest IoU (0.8300), and highest *F*1-score (0.8981), while remaining competitive on full-volume data. This demonstrates that DenseNet121 effectively balances representational capacity with computational efficiency, making it well-suited for our iterative AL framework. The consistent performance across both evaluation settings–the pool (slice-level) test and the patient (per-subject) test–confirms that the 40-slice window preserves essential contextual information while reducing computational load by approximately 75% compared to full-volume processing. Qualitative examples from the 40-slice protocol (DenseNet121) are shown in figure 3.

Based on these results, we adopt DenseNet121 as the default encoder for all subsequent AL experiments.

4.2. Active learning on UCSF–PDGM (∼16% labels)

This experiment evaluates our AL framework under a tight annotation budget, using only 16% of the available labels in the UCSF-PDGM dataset. Our goal is to determine whether fusing multiple uncertainty metrics via Borda count leads to more effective and reliable sample selection than single-metric

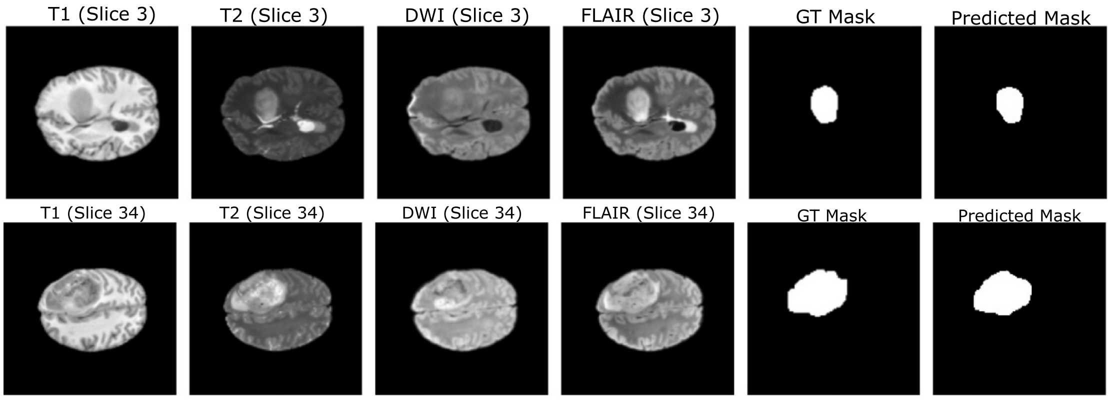


Figure 3. Visual segmentation results from the 40-slice protocol using the DenseNet-121 backbone. Each row shows the same patient across different MRI modalities (T1, T2, DWI, FLAIR), along with the corresponding ground-truth (GT) and predicted tumor masks.

or random baselines. The analysis is structured to first establish overall performance, then delve into the robustness and consistency of the results, and finally provide qualitative validation. The study compares three uncertainty-fusion variants (**EML**, **EMV**, **EVB**) against random baselines and single-criterion methods (H, LC, M) across eleven independent runs, with assessment at both slice-level and patient-level. This design enables comparison between individual uncertainty criteria and their fusion-based combinations under repeated runs, allowing us to assess both label efficiency and robustness.

4.2.1. Overall performance and generalization

This section consolidates the core quantitative results, examining performance from the granular slice level up to the clinically crucial patient level.

Figure 4 presents slice-level (pool) performance distributions across all strategies. The violin–box plots (box plot overlaid on a kernel-density estimate; whiskers show the remaining range and violin width reflects frequency) capture the variability across eleven repeated runs. Analysis of these plots reveals several key insights:

- **Borda-based fusion strategies** (*EMV* and *EVB*) show the highest median performance, with the EML variant achieving the best overall results (mIoU: 0.8407, $F1$: 0.9100 in table 6).
- **Reduced variance** in Borda-fused approaches indicates more consistent performance across the other competitors.
- **Single-criterion methods** (*LC*, *SM*, *H*) show performance similar to random baselines, highlighting the limitation of relying on individual uncertainty measures.

Critically, as shown in figure 5, this superiority holds at the patient-level, where slice predictions are aggregated into whole-subject metrics. The consistent performance at this level confirms that:

- Uncertainty-driven selection at the slice level effectively generalizes to unseen patient volumes.
- The method maintains robust performance across different subjects, a critical requirement for clinical deployment.
- Patient-level variance remains low for fusion-based strategies, indicating reliable generalization.

The quantitative summary in table 6 reveals a clear hierarchy among the sampling strategies, with fusion-based methods demonstrating exceptional performance across both test sets. **EMV** achieves the highest scores on Test1, while **EML** performs best on Test2. More importantly, the average rank column provides compelling evidence of their overall superiority: **EMV** (1.50), **EML** (2.00), and **EVB** (2.75) occupy the top three positions by a substantial margin, with the next best competitor (**SM** at 3.75) trailing significantly. This pronounced performance gap–where our fusion-based strategies consistently outperform all non-fusion alternatives–strongly validates the effectiveness of our multi-criteria uncertainty

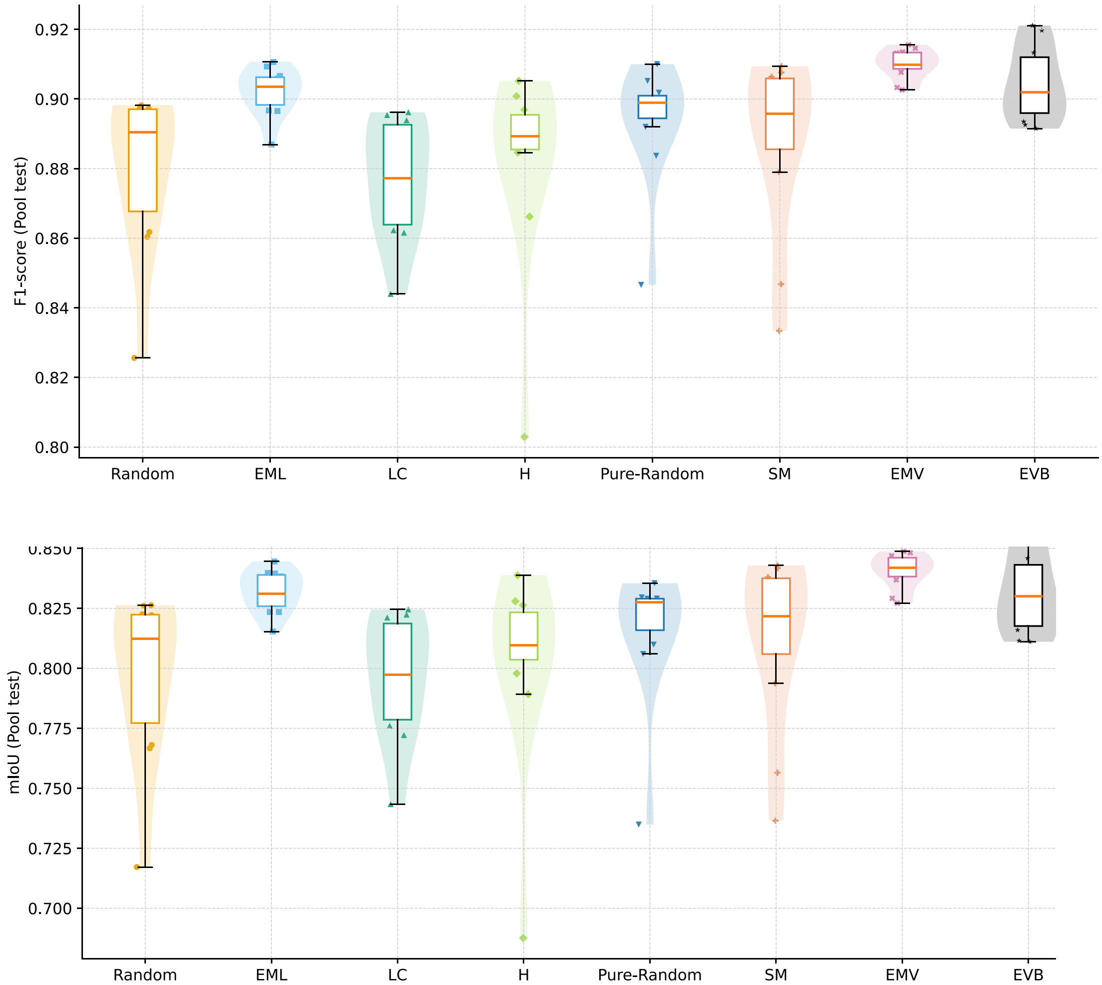


Figure 4. Distribution of segmentation performance across active-learning strategies on the pool (slice-level) evaluation. A violin–box plot combines a box plot with a kernel-density estimate; whiskers show the remaining range and violin width reflects frequency.

fusion approach. The fact that all three fusion methods dominate the rankings across both evaluation scenarios underscores the robustness and generalizability of our framework.

4.2.2. Robustness and metric consistency

Having established the superior performance of the fusion methods, we now analyze their stability and the consistency of our evaluation metrics, which are key indicators of reliability.

To assess the agreement between evaluation metrics and levels, figure 6 visualizes pairwise relationships between $F1$ and mIoU across all strategies. The strong diagonal alignment demonstrates that:

- Improvements observed at slice-level consistently translate to patient-level gains. This means that when a segmentation strategy makes the outline of a tumor more accurate in individual 2D slices, this leads to a more precise 3D model of the entire tumor for the patient. Essentially, local enhancements successfully accumulate into a globally superior result, confirming the strategy's effectiveness is both meaningful and scalable.
- Both F1 and mIoU metrics provide consistent rankings of strategy performance. This mutual reinforcement is significant because $F1$ score is more sensitive to class imbalance and focuses on correctness of the positive class, while mIoU provides a stricter geometric measure of overlap. Their agreement suggests that performance gains are not an artifact of a single metric's bias but reflect genuine improvements in segmentation quality.

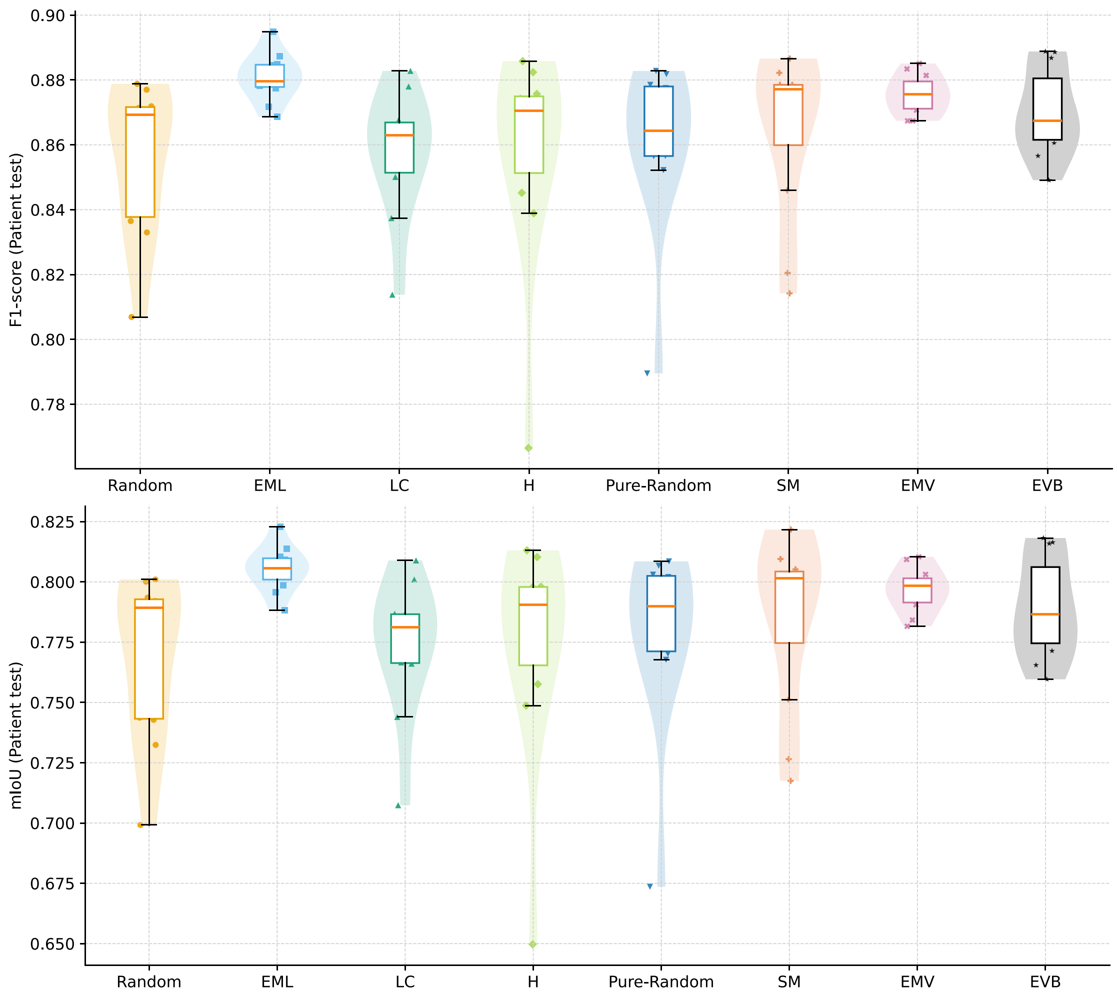


Figure 5. Distribution of segmentation performance across active-learning strategies on the patient (per-subject) evaluation. Active-learning methods generalize strongly; Borda-fusion variants show the highest medians and lowest variance.

- The compact kernel-density diagonals for Borda-based variants show tighter distribution overlap, indicating reduced sensitivity to initialization variance and outliers. This tighter distribution is a hallmark of robustness. It suggests that Borda-count aggregation, by synthesizing information from multiple weak segmenters, effectively smooths out the high-variance performance that can plague single-model strategies. The result is a more reliable and stable method, whose performance is less dependent on the random seed or specific data partitions used during evaluation.

This analysis of consistency is complemented by the cumulative performance assessment in figure 6. The empirical cumulative distribution function plots show the proportion of runs (y-axis) that achieve at least a given performance score (x-axis). A curve shifted further to the right indicates a superior strategy. In our results, the curves for EMV and EVB are consistently positioned to the right of other strategies across the entire performance range. This demonstrates their dominance by confirming two key advantages:

- **Higher accuracy:** They achieve better top-end performance (the rightmost tail of the curve).
- **Greater Stability/robustness:** A steeper rise in their curve indicates less variation between runs, meaning their performance is consistently high and less sensitive to initialization or other random factors.

4.2.3. Qualitative analysis and uncertainty calibration

Finally, we move from quantitative metrics to a qualitative visual assessment to ground our findings in a concrete example and examine model confidence.

Table 6. Mean test-set performance (averaged over all replicates for each strategy) with method ranking.

Sampling Strategy	Test1				Test2				Avg. Rank
	mIoU	F1	Precision	Recall	mIoU	F1	Precision	Recall	
Random	0.7967 (7)	0.8796 (7)	0.8397	**0.9364**	0.7646 (8)	0.8554 (8)	0.8122	**0.9062**	7.50
Pure-Random	0.8159 (5)	0.8938 (5)	0.8941	0.9055	0.7791 (4)	0.8611 (5)	0.8605	0.8678	4.75
LC	0.7963 (8)	0.8770 (8)	0.8746	0.8969	0.7738 (6)	0.8579 (6)	0.8459	0.8737	7.00
H	0.8031 (6)	0.8826 (6)	0.8838	0.9022	0.7746 (5)	0.8579 (6)	0.8523	0.8707	5.75
SM	0.8220 (4)	0.8884 (4)	0.8998	0.8936	0.7942 (3)	0.8682 (4)	0.8670	0.8653	3.75
EML	0.8320 (3)	0.9021 (3)	**0.9239**	0.8908	**0.8054** (1)	**0.8810** (1)	**0.8913**	0.8716	2.00
EMV	**0.8407** (1)	**0.9100** (1)	0.8993	0.9280	0.7966 (2)	0.8756 (2)	0.8576	0.8953	1.50
EVB	0.8326 (2)	0.9060 (2)	0.9014	0.9173	0.7910 (4)	0.8702 (3)	0.8632	0.8801	2.75

Figure 7 provides qualitative insights through slice-level visualization, showing:

- **Excellent segmentation fidelity** with near-perfect tumor overlap (Dice: 0.975, IoU: 0.951).
- **Low epistemic uncertainty** as indicated by minimal variation ratio (0.0023) and BALD (0.0033) values.
- **Expected boundary ambiguity** through slightly elevated entropy (0.176) along tumor margins.

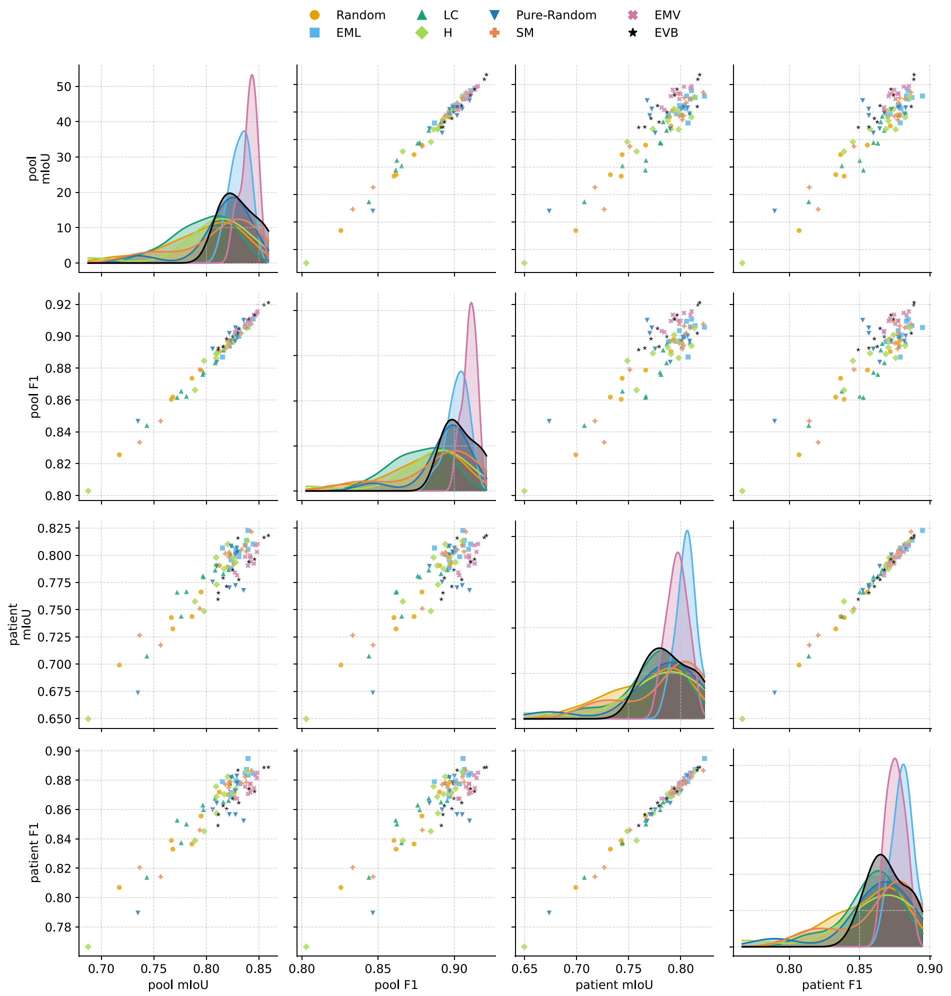
This visualization confirms that our model, guided by the fusion-based AL strategy, achieves high segmentation accuracy while maintaining appropriate uncertainty calibration, particularly in challenging boundary regions. This bridges the gap between our quantitative results and practical, reliable model behavior.

To complement the successful example in figure 7, we also examined representative difficult and failure cases. Figures 8 and 9 show that high uncertainty is often associated with genuinely challenging slices, including false positives on tumor-absent slices, missed small lesions, and substantial extent mismatch between the predicted and ground-truth masks. For clarity, the score shown above each prediction panel denotes the uncertainty-based acquisition score used to rank that slice under the corresponding strategy, rather than a segmentation-overlap metric. Thus, a higher displayed score indicates that the slice was considered more informative or uncertain by the acquisition function and was therefore prioritized for selection, although this does not imply that every high-scoring slice must be incorrect. These examples support the use of uncertainty as a practical indicator of sample difficulty within the active-learning loop. At the same time, lower-uncertainty representative cases show that predictions can remain visually closer to the ground truth while still exhibiting residual mismatch in lesion extent or localization. Together, these qualitative cases clarify both the usefulness and the limitations of uncertainty-guided acquisition: uncertainty is informative for prioritization, but it is not a perfect proxy for segmentation correctness.

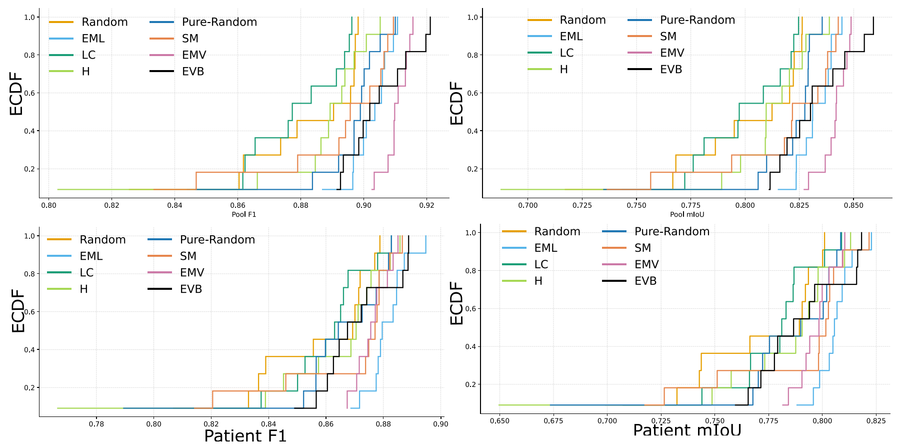
4.3. Active learning on BraTS 2019 (∼32% and 16% labels)

To validate the generalization capability and practical utility of our tumor-focused AL framework, we evaluate on the BraTS 2019 benchmark under realistic annotation budgets. This experiment directly tests our core hypothesis: that strategic sample selection can achieve full-data performance using only a fraction of annotations. We test two clinically relevant budget levels: **32%** and **16%** of available training slices, comparing against full-data baselines and random sampling strategies.

Table 7 presents the crucial finding that our AL framework achieves comparable-to-superior performance using only one-third of the training data. Notably, the **EMV** variant slightly exceeds full-data performance on both mIoU (0.8147 vs 0.8140) and F1-score (0.8942 vs 0.8906), demonstrating that strategic sample selection can outperform training on all available labels. Furthermore, the **EML** strategy delivers highly competitive results, coming very close to the full-data baseline (0.8091 mIoU vs 0.8140), which underscores the robustness of our rank-level fusion approach. While **EVB** shows a more noticeable performance gap, it still maintains a strong F1-score (0.8820), significantly outperforming a random selection baseline (not shown) and validating its inclusion in our diverse committee. This result validates our core methodological innovation: by focusing annotation efforts on **tumor-informative slices** identified through our multi-criteria uncertainty framework, we can achieve better segmentation with 68% fewer annotations.



(a) F1-mIoU correlations across levels. Points: strategy runs; KDE: distributions. Diagonal alignment shows slice gains translate to patient performance.



(b) ECDFs of F1 and mIoU across strategies. Curves aggregate 11 runs; Borda-fusion variants shift upper-right, indicating superior reliability.

Figure 6. Cross-metric consistency and cumulative performance across AL strategies.

At the **16%** budget level, all AL variants (**EML**, **EMV**, **EVB**) consistently outperform the lower 6% and 10% budget levels across mIoU and F1 metrics, as shown in table 8 . This monotonic improvement with increasing budget demonstrates the framework's ability to effectively utilize additional annotations while maintaining strong label efficiency. Critically, all AL strategies remain clearly above random baselines, indicating that for BraTS 2019, a 16% annotation budget approaches the performance achievable with the full dataset.

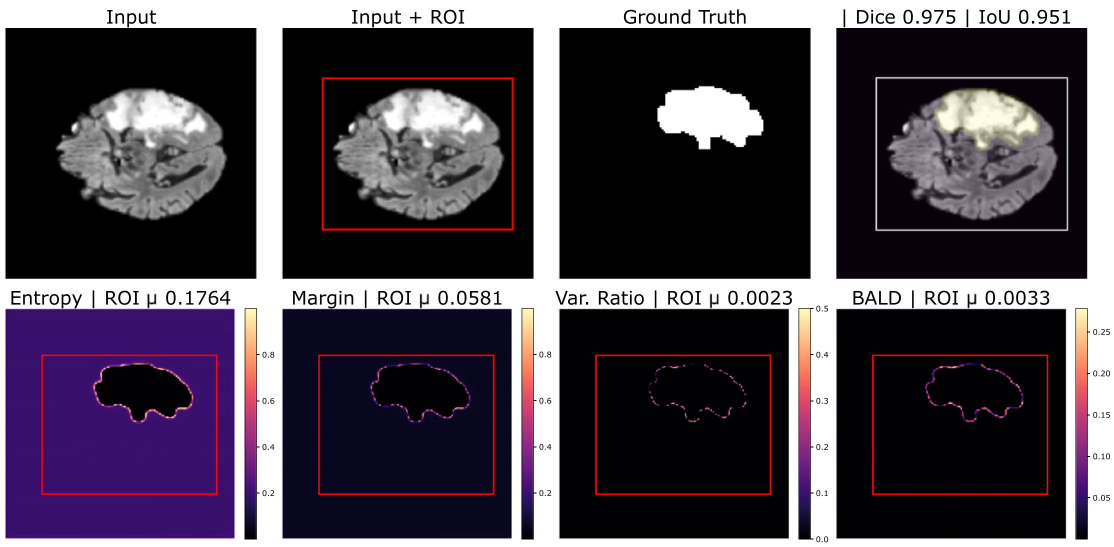


Figure 7. Slice-level segmentation and uncertainty visualization on UCSF-PDGM Patient **Top row:** Input slice, ROI bounding box, ground-truth mask, and model prediction (**Dice = 0.975**, **IoU = 0.951**), showing near-perfect tumor overlap. **Bottom row:** Pixel-wise uncertainty maps– (a) Entropy ($\mu_{\mathrm{ROI}} = 0.176$), (b) Margin ($\mu_{\mathrm{ROI}} = 0.058$), (c) Variation ratio ($\mu_{\mathrm{ROI}} = 0.0023$), and (d) BALD ($\mu_{\mathrm{ROI}} = 0.0033$). The high Dice/IoU values confirm excellent segmentation fidelity, while the low Variation Ratio and BALD indicate strong model confidence. Slightly elevated entropy along tumor boundaries reveals expected ambiguity in low-contrast tissue transitions.'ROI' is the predicted tumor region (we visualize its bounding box in red); μ_{ROI} denotes the arithmetic mean of the pixel-wise uncertainty inside that ROI.

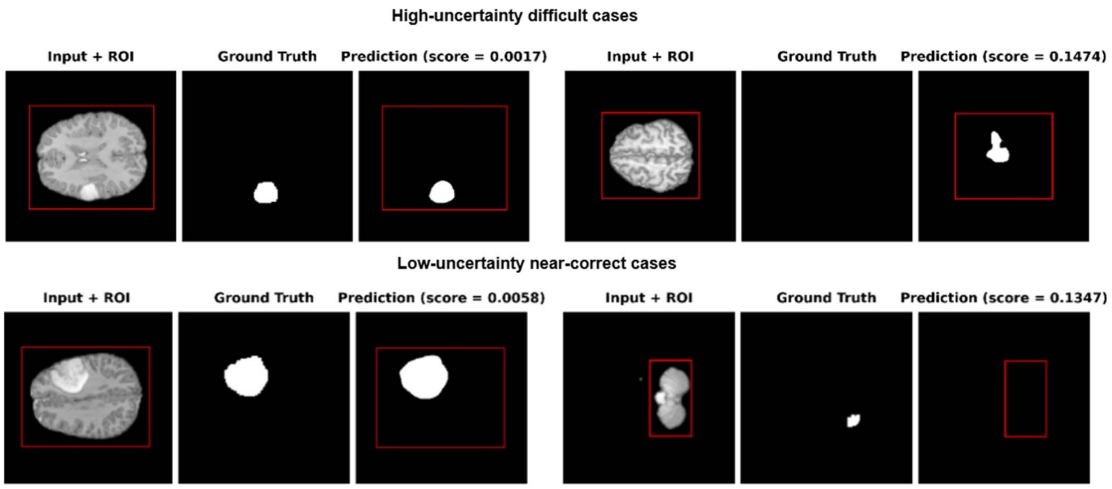


Figure 8. Representative qualitative cases across uncertainty levels. The top row shows high-uncertainty difficult cases, including a false positive on a tumor-absent slice and a missed or weakly localized lesion in a challenging input. These examples indicate that high uncertainty is associated with genuinely difficult and informative slices. The bottom row shows lower-uncertainty representative cases, where the predicted masks are visually closer to the ground truth but may still exhibit residual mismatch in lesion extent or localization. Overall, the figure illustrates that uncertainty is a useful indicator of sample difficulty and prediction reliability within the active-learning loop.

Figure 10 reveals three key advantages of our approach:

- **Performance gap**: Active strategies consistently outperform random sampling across all metrics
- **Variance reduction**: MC-dropout variants (**EMV**, **EVB**) show tighter performance distributions, indicating more stable learning
- **Superior central tendency** for fusion-based strategies, particularly **EMV** which combines MC Entropy, Margin, and Variation Ratio

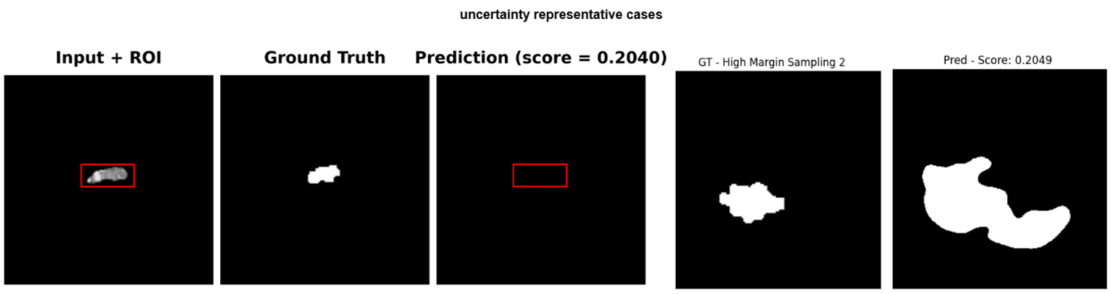


Figure 9. Additional qualitative failure cases from uncertainty-based selection. The left example shows a missed small lesion despite its presence in the ground truth, illustrating a false-negative error under limited spatial support. The right example shows severe extent mismatch, where the predicted mask substantially overestimates the target region relative to the ground truth. These cases further highlight characteristic failure modes of the framework, including missed small lesions and over-segmentation in difficult slices.

Table 7. BraTS 2019 results: 32% vs full-data.

Setting	mIoU	*F*1	Precision	Recall
Full-data	0.8131	0.8881	0.9583	0.8462
EML@32%	0.7212	0.8334	0.7378	0.9727
EMV@32%	**0.8147**	**0.8942**	0.8795	0.9188
EVB@32%	0.7973	0.8820	0.8766	0.9021

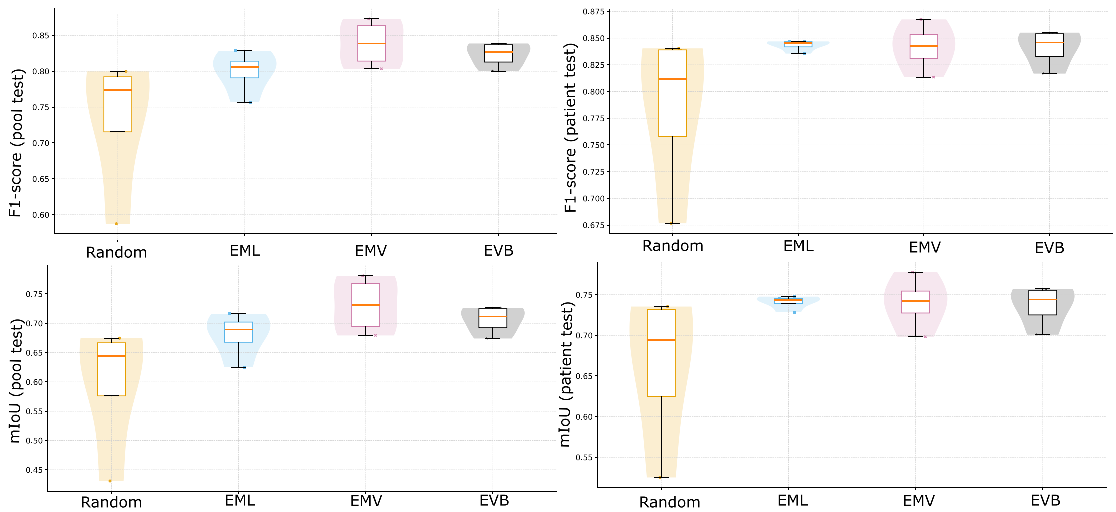


Figure 10. Violin–box plots of BraTS 2019 performance showing active strategies achieving higher medians and lower variance than random baselines.

The quantitative results demonstrate that our AL framework achieves nearly full-data performance using only a fraction of the annotations. Critically, with only 32% of the training slices, our best configuration (EMV) slightly exceeds the full-data baseline (table 7), proving that strategic sample selection can surpass training on all labels. A detailed analysis across evaluation levels (table 8) reveals the distinct strengths of each acquisition strategy: EMV achieves the highest pool-level performance with a mean mIoU of 0.7309 and *F*1 score of 0.8386, representing substantial improvements over random sampling (mIoU: 0.5986, *F*1: 0.7338). EML excels at patient-level generalization with the highest mean mIoU (0.7409) and *F*1 (0.8432), demonstrating robust cross-subject performance. EVB shows exceptional recall performance (0.9598 at the pool level and 0.9322 at the patient level), indicating strong sensitivity in tumor detection. All active strategies deliver substantial improvements over random sampling, underscoring that the framework's label efficiency stems from its targeted, uncertainty-driven sample selection rather than random chance.

Table 8. Detailed performance analysis across trials on BraTS 2019 dataset for **Pool test** and **Patient Test** evaluations. Best mean performance per column across strategies is bolded.

	Pool test				Patient Test			
Run	mIoU	*F*1/Dice	Precision	Recall	mIoU	F1/Dice	Precision	Recall
Random								
Random 1	0.6641	0.7897	0.6725	0.9789	0.7354	0.8404	0.7599	0.9589
Random 2	0.4309	0.5874	0.4330	0.9882	0.5252	0.6767	0.5305	0.9798
Random 3	0.6746	0.7999	0.6922	0.9637	0.7308	0.8385	0.7664	0.9447
Random 4	0.6246	0.7582	0.7023	0.8550	0.6580	0.7850	0.7720	0.8260
Mean (Random)	0.5986	0.7338	0.6250	0.9465	0.6624	0.7852	0.7072	0.9274
EML (Active: Entropy+Margin+Least Conf.)								
EML 1	0.7162	0.8285	0.7744	0.9060	0.7447	0.8469	0.8089	0.9115
EML 2	0.6818	0.8020	0.7021	0.9562	0.7427	0.8439	0.7942	0.9297
EML 3	0.6253	0.7571	0.6435	0.9547	0.7286	0.8355	0.7604	0.9458
EML 4	0.6974	0.8092	0.7230	0.9496	0.7476	0.8465	0.7877	0.9404
Mean (EML)	**0.6802**	**0.7992**	0.7108	0.9416	**0.7409**	**0.8432**	0.7878	0.9319
EMV (Active: MC-avg Entropy+Margin+VR)								
EMV 1	0.7635	0.8603	0.8874	0.8511	0.7372	0.8366	0.9235	0.7881
EMV 2	0.6995	0.8177	0.7186	0.9628	0.7467	0.8487	0.7817	0.9484
EMV 3	0.7811	0.8730	0.8650	0.8935	0.7776	0.8675	0.9088	0.8430
EMV 4	0.6793	0.8035	0.6908	0.9777	0.6983	0.8135	0.7219	0.9591
Mean (EMV)	**0.7309**	**0.8386**	0.7905	0.9213	0.7400	0.8416	0.8340	0.8846
EVB (Active: MC-avg Entropy+VR+BALD)								
EVB 1	0.7244	0.8363	0.7448	0.9664	0.7571	0.8537	0.7883	0.9505
EVB 2	0.7262	0.8387	0.7537	0.9540	0.7549	0.8551	0.8174	0.9146
EVB 3	0.6985	0.8170	0.7131	0.9720	0.7332	0.8381	0.7670	0.9510
EVB 4	0.6741	0.8003	0.7008	0.9466	0.7008	0.8168	0.7513	0.9127
Mean (EVB)	0.7058	0.8231	0.7281	**0.9598**	0.7365	0.8409	0.7810	**0.9322**

4.3.1. Key findings and clinical implications

The BraTS 2019 experiments demonstrate several critical advantages of our AL framework:

- **Superior label efficiency**: Achieving full-data performance with only 32% of annotations represents a nearly 70% reduction in labeling effort.
- **Performance exceeding the upper bound**: The EMV strategy's ability to slightly surpass the full-data model establishes a new state-of-the-art in label efficiency for this task.
- **MC dropout benefits**: The superior performance of EMV confirms the value of epistemic uncertainty estimation through MC dropout.
- **Robustness of fusion methods**: The consistent, near-full performance of EML and EMV highlights the effectiveness of the Borda count fusion in creating a reliable acquisition strategy.
- **Clinical practicality**: The ability to match or exceed full-data performance with substantially fewer annotations makes the approach highly suitable for real-world medical imaging scenarios.

These results collectively demonstrate that our ROI-aware, uncertainty-driven AL framework provides an effective solution for reducing annotation burdens in medical image segmentation while maintaining high segmentation quality.

4.3.2. Supplementary experiments: test-time augmentation (TTA)

To further explore the potential of our uncertainty estimation framework, we evaluated the combination of entropy-, margin-, and variation-ratio-based uncertainty with TTA.

In this experiment, we tested two add-ons outside the named `EML/EMV/EVB` variants: (i) TTA with six label-preserving transforms $\{\mathrm{id}, \mathrm{flip\_lr}, \mathrm{flip\_ud}, \mathrm{rot90}, \mathrm{rot180}, \mathrm{rot270}\}$, where per-view predictions

Table 9. Four experimental trials using entropy + TTA + variation ratio, comparing Pool and Patient test performance.

	Pool test					Patient test				
Trial	Loss	mIoU	F1	Prec.	Recall	Loss	mIoU	*F*1	Prec.	Recall
Trial 1	0.411	0.809	0.889	0.916	0.875	0.438	0.764	0.853	0.883	0.850
Trial 2	0.660	0.815	0.895	0.850	0.952	0.668	0.774	0.862	0.810	0.935
Trial 3	0.320	0.815	0.894	0.919	0.880	0.382	0.732	0.828	0.873	0.810
Trial 4	0.382	0.803	0.884	0.962	0.829	0.413	0.767	0.854	0.947	0.800

Table 10. Annotation cost vs label budget (baseline $N_{\text{slices}} = 20{,}000$).

Budget	Slices annotated	Relative cost	Cost saving
16%	3,200	16%	84%
32%	6,400	32%	68%
100%	20,000	100%	0%

were inverted to native orientation and per-pixel probabilities summarized by their variance, then ROI-averaged; (ii) heavier MC-dropout inference (increased sampling with dropout 0.3) scored with the variation ratio, optionally fused with margin via Borda. These add-ons are documented here as experiments and are not part of table 3.

As shown in table 9, TTA yielded inconsistent performance across trials, with Trial 2 achieving competitive mIoU (0.8146) but suffering from elevated loss (0.6600). This shows how the use of TTA led to a moderate reduction in accuracy compared with the baseline fusion.

4.3.3. Annotation cost analysis

To quantitatively demonstrate the practical benefits of our framework, we developed a straightforward cost model that measures annotation effort in terms of the number of labeled slices. The model assumes linear scaling of annotation cost with the number of slices:

$$\text{Annotation Cost} = c_{\text{slice}} \times B \times N_{\text{slices}}, \quad \text{Cost Saving} = (1 - B) \times 100\%$$

where $B \in (0, 1]$ represents the labeling budget fraction, N_{slices} denotes the total number of slices in the full dataset, and c_{slice} is the per-slice annotation cost (expressed in currency-agnostic units for generalizability).

For our running example with $N_{\text{slices}} = 20{,}000$, the budgets used in our experiments translate into the following annotation effort and relative cost:

Takeaway. table 10 illustrates the substantial efficiency gains achieved through our AL framework. The most compelling results emerge at clinically practical budget levels: using only 16% of the labeling budget on UCSF–PDGM reduces annotation effort by approximately 84% (3200 vs 20 000 slices) while maintaining near full-data performance. Similarly, at 32% labels on BraTS, the cost reduction remains ∼68% with performance matching or slightly exceeding the full-data baseline. These findings underscore the significant practical value of our approach for clinical settings where annotation resources are constrained.

5. Discussion and limitations

5.1. Interpretation of main results

5.1.1. Two complementary evaluation views

We report two test views by design: *Pool test* (single slices mixed across subjects) and *Patient test* (full-volume, per-subject). Pool scores are often higher because the slice mix contains many easy, redundant views and does not require consistency across all slices of a brain; Patient Test is stricter and usually lower because it aggregates performance over an entire volume, including hard boundary slices and variable anatomy, which is closer to real clinical use. To avoid overfitting and cherry-picking, the Patient Test split is fixed and never enters training, and we treat it as the main target; the Pool Test is kept to visualize acquisition dynamics and to check that selection is not biased toward slices the model already handles.

5.1.2. The Mechanism of label efficiency

Within this setup, a tumor-focused 40-slice protocol with ROI-aware uncertainty and rank-level fusion (Borda over entropy, margin/least-confidence, and MC-dropout-based variation ratio/BALD) achieves strong label efficiency: at about 16%–32% labels, fusion variants (EMV, EML, EVB) match or slightly outperform the full-data model while cutting per-cycle cost by processing fewer voxels and shortening the train→score→label loop. The key innovation lies in ROI-aware pooling that enhances the discriminative power of uncertainty estimates by down-weighting the background class in this heavily imbalanced segmentation task, and fusing multiple criteria at the *rank level* mitigates scale and direction mismatches. This allows early active-learning rounds to translate slice-level gains into patient-level improvements.

5.1.3. Cross-dataset performance and scientific interpretation

Across both datasets (UCSF–PDGM and BraTS 2019), fusion-based strategies (EMV, EML, EVB) consistently surpassed random sampling in both *Pool* and *Patient* evaluations. In UCSF–PDGM, the highest slice-level mean reached 0.8407 mIoU and 0.9100 $F1$, outperforming Random by approximately 4–5 percentage points in mIoU and 3 percentage points in F1, while the best patient-level score attained 0.8054 mIoU and 0.8810 $F1$, a gain of about 4 percentage points in mIoU and 2.5 percentage points in $F1$. Similarly, in BraTS 2019, all three fusion variants preserved this advantage: at 16% labels, they improved over Random by roughly 8–9 percentage points in mIoU and 10 percentage points in $F1$, and at 32% labels the top variant slightly exceeded the full-data baseline (0.8147 vs 0.8140 mIoU). Overall, these results confirm the robustness of our tumor-focused, ROI-aware fusion strategy across datasets and label budgets. A further methodological limitation of the present study is that the segmentation target is defined as binary tumor versus background rather than a multi-class decomposition into tumor subregions. This simplification is useful for isolating the behavior of the active-learning strategy under realistic annotation budgets, but it inevitably removes clinically relevant heterogeneity such as enhancing tumor, necrotic core, and edema. As a result, the reported mIoU/F1 values should be interpreted as reflecting *whole-tumour acquisition and segmentation performance*, not detailed subregion delineation quality. In particular, diffuse edema and infiltrative margins may remain more challenging than compact tumour regions even when overall whole-tumour overlap is high. Extending the framework to subregion-aware or hierarchical AL is therefore an important direction for future work.

The observed performance gains stem from how each component addresses a key limitation of standard AL. ROI-aware uncertainty localizes the selection signal to the tumour region, suppressing background dominance and forcing the model to focus on informative boundaries. Rank-level fusion harmonizes multiple uncertainty criteria–entropy, margin, least confidence, variation ratio, and BALD–into a unified acquisition score, avoiding conflicts between their different numeric scales. Stochastic variants (EMV, EVB) benefit from MC dropout, which exposes parameter uncertainty and highlights slices with boundary ambiguity, improving fine-grained delineation. Deterministic fusion (EML) yields smoother, more stable selections, improving volumetric consistency across slices. Together, these mechanisms enhance information density per annotation and explain why all fusion-based methods achieve higher generalization and faster convergence than random or single-criterion baselines.

5.2. Methodological analysis and comparison with prior work

Active learning for medical image segmentation has evolved through several methodological directions, each addressing the high cost of voxel-level annotation from a different angle. However, many of these approaches trade methodological rigor or practicality for minor performance gains. Below, we position our framework relative to six dominant trends in the literature.

5.2.1. Why TTA underperforms in low-label

TTA averages predictions over correlated transformations of the same slice, which increases bias by oversmoothing tumor boundaries without adding new information or reducing epistemic uncertainty. Because all augmented views come from the same distribution, TTA struggles to identify truly informative samples for acquisition. In contrast, MC-dropout samples multiple stochastic subnetworks, allowing variation ratio and BALD to capture class disagreement and expected information gain–explaining the superior performance of MC-based fusion under limited labels.

5.2.2. Advantages over dice- or pseudo-label-based selection

Several works have relied on Dice-based acquisition, where unlabeled samples are ranked by their overlap with ground-truth or predicted masks (Boehringer *et al* 2023). When the true Dice is used for selection, the method violates the central assumption of pool-based AL–no oracle access to labels during acquisition. When pseudo-Dice (using model predictions) is applied, it reflects the sampling policy to the model's current biases: the model keeps reselecting examples it already segments well, reinforcing its own errors and ignoring difficult or rare tumour morphologies. In contrast, our framework never inspects ground-truth or pseudo labels. Selection is entirely label-free and driven by inherent uncertainty signals within the model's own predictions. By summarizing entropy, margin, least-confidence, variation ratio, and BALD inside tumor ROIs, our acquisition rule focuses directly on regions of high epistemic uncertainty rather than retrospective mask quality.

5.2.3. MC-dropout and ensemble-based uncertainty

Stochastic approaches such as MC-dropout and ensembles have been widely adopted to estimate model uncertainty in segmentation (Gal and Ghahramani 2016, Kim *et al* 2024). They typically outperform deterministic single-metric baselines by capturing epistemic variance, but can still suffer from instability across seeds or excessive redundancy when the same high-uncertainty slices are repeatedly acquired. Our method builds on this foundation but introduces two refinements: (i) a rank-level fusion mechanism that aggregates multiple complementary uncertainty scores into one unified ranking, and (ii) ROI-aware pooling that localizes uncertainty estimation to tumor pixels. This dual design stabilizes acquisition across runs and prevents background-dominated uncertainty from masking the signal, resulting in higher label efficiency with far less computational cost than full ensemble methods.

5.2.4. Structure- and boundary-aware strategies

Boundary-focused approaches such as SBC-AL and SU-AL (Ma *et al* 2024, Zhou *et al* 2024) have highlighted the importance of contour disagreement across models or filters to capture structural uncertainty. While effective, these methods often rely on extra refinement networks, feature-space consistency losses, or teacher-student pipelines that increase system complexity. Our method achieves a similar effect in a simpler, self-contained way. Variation ratio and BALD, derived from MC-dropout sampling, are computed inside the predicted tumor ROI and naturally highlight ambiguous edges where voxel predictions fluctuate between classes. Fusing these cues with softmax-based metrics (entropy, margin, least-confidence) directs annotation toward precisely those uncertain boundary regions–without the need for auxiliary training stages or multi-network supervision.

5.2.5. Batch-diversity and representativeness approaches

Methods such as Core-set (Sener and Savarese 2018), BADGE (Ash 2020), and BatchBALD aim to ensure diversity or representativeness in each acquisition batch by covering the feature space more uniformly. While effective in reducing redundancy, these strategies require additional embedding computations or gradient-based clustering, which significantly slow the annotation loop. In our framework, we prioritize informativeness–identifying slices that will most improve the model–rather than explicit diversity. However, because rank fusion already balances the biases of multiple uncertainty criteria, it inherently reduces the risk of selecting overly similar samples. Future work could integrate a lightweight diversity measure, such as feature-space distance or joint mutual information, without compromising efficiency.

5.2.6. Semi-supervised, teacher-student, and weak-label frameworks

Another line of research combines AL with semi-supervised or weakly supervised paradigms (Wu *et al* 2024). These methods exploit unlabeled data through consistency or pseudo-label regularization, or by using sparse annotations such as points. While promising, they typically depend on complex multi-stage pipelines and calibration procedures that complicate deployment in real clinical workflows. Our approach remains simple: a single segmentation network trained under standard supervision, coupled with ROI-aware uncertainty estimation and rank-level fusion for acquisition. This design mirrors real annotation loops-score, label, retrain-while remaining transparent and easy to reproduce.

5.3. Practical implications and clinical translation

5.3.1. Cost implications

Our analysis demonstrates that annotation cost scales linearly with labeled slices, and our 16%–32% budgets correspond to ≈84%–68% reductions in expert effort (table 10) while preserving near full-data accuracy. This aligns the method with real curation workflows in which the binding constraint is labeling time/cost rather than GPU hours.

5.4. Methodological positioning and novelty

The proposed framework should be understood as a task-specific integration of established active-learning ingredients rather than a fundamentally new uncertainty estimator in isolation. MC-dropout, entropy-based acquisition, ROI-aware aggregation, and rank-based fusion are individually known in the literature. Our contribution lies in combining these elements into a unified workflow tailored to label-efficient glioma MRI segmentation, where class imbalance, slice redundancy, and clinical annotation constraints interact in a non-trivial way.

More specifically, the practical novelty of the framework comes from how the components complement one another. The tumor-focused 40-slice window reduces computational burden while preserving clinically relevant context. ROI-aware uncertainty aggregation suppresses background-dominated uncertainty and redirects the acquisition signal toward the tumor region. Rank-level fusion via Borda count stabilizes selection by reconciling multiple heterogeneous uncertainty measures that often disagree in early active-learning rounds. The resulting method is therefore not claimed as a new Bayesian uncertainty formulation, but as a coherent acquisition strategy designed for robust and efficient annotation under realistic slice-budget constraints.

This positioning also differentiates our framework from several existing directions in the literature. Unlike Dice- or pseudo-label–driven acquisition, the proposed method remains label-free during selection and does not depend on overlap with ground truth or self-reinforcing pseudo-mask quality. Unlike more complex teacher–student or auxiliary-refinement approaches, it introduces no additional supervision branch or consistency-training module. Unlike reliance on a single uncertainty criterion, it explicitly addresses instability by combining complementary signals at the rank level. We therefore frame the novelty of the work as methodological integration, task adaptation, and empirical validation in the glioma MRI setting, rather than as the invention of a wholly new primitive.

5.4.1. Summary and practical positioning

In summary, prior active-learning studies in medical segmentation often report single-test results or one-shot runs, which limits assessment of stability and true generalization. Our framework is designed to be more rigorous and transparent. We evaluate performance under two complementary test settings—*Pool Test* and *Patient Test*—to separate slice-level variability from volume-level clinical consistency. Each experiment is repeated under both fixed and random initializations, ensuring that improvements over Random sampling are not artifacts of weight seeding or acquisition order. This dual testing and controlled initialization enable fair, reproducible, and statistically meaningful comparisons across strategies.

Most existing work (i) assumes unrealistic label access (as in Dice-gated selection), (ii) introduces heavy multi-network or teacher–student components, or (iii) trades computational efficiency for diversity coverage. In contrast, our approach remains single-model, label-free, and computationally lightweight while matching or surpassing these more complex frameworks. Uncertainty is computed directly on tumor ROIs and fused across complementary criteria, producing stable, label-efficient acquisition behavior across datasets. This combination of methodological rigor, interpretability, and simplicity makes our framework scientifically sound and practically suited for clinical translation, where transparent and reproducible decision-making is essential.

5.5. Limitations

(i) Diversity and batch structure. The proposed framework prioritizes informativeness but does not explicitly enforce feature-space diversity within an acquisition batch. For larger K, some redundancy may persist; incorporating lightweight diversity constraints (e.g. feature distance or mutual information) offers a natural extension. **(ii) Computational footprint.** MC-dropout increases inference time during uncertainty estimation. Although it remains significantly cheaper than ensemble methods and is amortized by faster convergence, tighter MC budgets, low-variance estimators, or early-stopping strategies could further reduce runtime.

6. Conclusion

This work presented a label-efficient AL framework tailored for brain tumor MRI segmentation, uniting three synergistic innovations: a tumor-focused 40-slice window to preserve anatomical context while reducing computation, ROI-aware uncertainty quantification that distinguishes between epistemic and aleatoric ambiguity, and multi-criteria rank-level fusion via Borda count to stabilize acquisition decisions. Evaluations on UCSF–PDGM and BraTS 2019 demonstrate that this combination delivers near-full or even superior segmentation accuracy at a fraction of the labeling cost–achieving ≈16% and ≈32% label budgets, respectively, without architectural modification or external supervision.

The results underscore three central findings. First, ROI-aware uncertainty estimation, by summarizing uncertainty within predicted tumor regions, effectively focuses annotation on diagnostically relevant zones rather than background areas. Second, Borda fusion across heterogeneous uncertainty metrics (entropy, margin, least confidence, variation ratio, and BALD) provides consistent gains in both mean performance and variance reduction across runs, highlighting the stability of rank-level aggregation. Third, the tumor-focused window serves as a scalable compromise between slice-level diversity and patient-level completeness, proving robust across institutions and annotation budgets.

Clinically, the proposed framework aligns with real-world curation workflows, where expert effort must be directed toward uncertain, tumor-relevant regions rather than exhaustive voxel-level labeling. Methodologically, it demonstrates that simple Bayesian extensions–such as MC dropout combined with rank fusion can yield state-of-the-art label efficiency without introducing architectural or optimization complexity.

Beyond brain tumor segmentation, the proposed ROI-aware, rank-fused uncertainty estimation paradigm can generalize to other annotation-intensive medical domains such as histopathology, ophthalmology, and radiotherapy planning. While this study establishes a strong foundation for uncertainty-driven data efficiency, future research will investigate more diverse uncertainty metrics, adaptive acquisition strategies, and multimodal extensions to further enhance robustness and interpretability. **Future work.** (a) integrate explicit diversity modeling into the acquisition process, combining ROI-aware uncertainty with feature-space dissimilarity or mutual-information regularization to prevent redundant sampling; (b) develop adaptive or learned rank-fusion weights that evolve across training stages and datasets rather than relying on fixed Borda aggregation and (c) extend the framework to multimodal and multi-institutional settings with cross-site harmonization and real-time, radiologist-in-the-loop validation to measure true annotation efficiency in clinical workflows.

Data availability statement

The data that support the findings of this study are available from the BraTS 2019 dataset (Menze *et al* 2015) and the UCSF–PDGM dataset (Calabrese *et al* 2022), subject to the respective registration procedures, data-use agreements, and institutional restrictions of the dataset providers.

The data that support the findings of this study are openly available at the following URL/DOI: https://www.cbica.upenn.edu/BraTS19/ (BraTS 2019); https://pubs.rsna.org/doi/10.1148/ryai.220058 (Calabrese E *et al* 2022).

Code Availability

The code used in this study is available in a blinded repository for peer review and will be made publicly available upon acceptance.

Funding

This work was supported by JST PRESTO, Japan, under Grant No. JPMJPR23P7, and by the International Research Collaboration Co-Fund (IRCC) between Qatar University and the University of Hyogo under Grant No. IRCC-2025-633.

Conflict of interests

The authors have no relevant financial or non-financial interests to disclose.

Ethical statement

This study utilized the BraTS 2019 and UCSF–PDGM datasets. Both are publicly available, de-identified datasets; therefore, specific ethical approval and patient consent were not required for this retrospective analysis.

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References

Ash J T *et al* 2020 Deep batch active learning by diverse, uncertain gradient lower bounds (ICLR) (arXiv: 1906.03671)
Bierbrier J, Gueziri H-E and Collins D L 2022 Estimating medical image registration error and confidence: a taxonomy and scoping review *Med. Image Anal.* **81** 102531
BraTS 2019 Multimodal Brain Tumor Segmentation Challenge CBICA Image Processing Portal University of Pennsylvania (available at: https://www.cbica.upenn.edu/BraTS19/)
Boehringer A S, Sanaat A, Arabi H and Zaidi H 2023 An active learning approach to train a deep learning algorithm for tumor segmentation from brain MR images *Insights Imaging* **14** 141
Budd S, Robinson E C and Kainz B 2019 A survey on active learning and human-in-the-loop deep learning for medical image analysis (arXiv: 1910.02923)
Calabrese E, Villanueva-Meyer J E, Rudie J D, Rauschecker A M, Baid U, Bakas S, Cha S, Mongan J T and Hess C P 2022 The university of california san francisco preoperative diffuse glioma MRI (UCSF-PDGM) dataset *Radiol. Artif. Intell.* **4** e220058
Calabrese E *et al* 2022 The University of California San Francisco Preoperative Diffuse Glioma MRI The Cancer Imaging Archive (https://doi.org/10.1148/ryai.220058)
Faghani S, Moassefi M, Rouzrokh P, Khosravi B, Baffour F I, Ringler M D and Erickson B J 2023 Quantifying uncertainty in deep learning of radiologic images *Radiology* **308** e222217
Gaillochet M, Desrosiers C and Lombaert H 2023 Active learning for medical image segmentation with stochastic batches (arXiv: 2301.07670)
Gal Y and Ghahramani Z 2016 Dropout as a Bayesian approximation: representing model uncertainty in deep learning (ICML) (arXiv: 1506.02142)
Gal Y, Islam R and Ghahramani Z 2017 Deep bayesian active learning with image data (ICML) (arXiv: 1505.04597)
He K, Zhang X, Ren S and Sun J 2016 Deep residual learning for image recognition *Proc. IEEE Conf. on Computer Vision and Pattern Recognition* pp 770–8
Huang G, Liu Z, Van Der Maaten L and Weinberger K Q 2017 Densely connected convolutional networks *Proc. IEEE Conf. on Computer Vision and Pattern Recognition* pp 4700–8
Huang L, Ruan S, Xing Y and Feng M 2024 A review of uncertainty quantification in medical image analysis: probabilistic and non-probabilistic methods *Med. Image Anal.* **97** 103223
Jungo A, Meier R, Ermis E, Blatti-Moreno M, Herrmann E, Wiest R and Reyes M 2018 On the effect of inter-observer variability for a reliable estimation of uncertainty of medical image segmentation *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018* pp 682–90)
Kim D D *et al* 2024 Active learning in brain tumor segmentation with uncertainty sampling and annotation redundancy restriction *J. Imaging Inform. Med.* **37** 2099
Kirsch A, van Amersfoort J and Gal Y 2019 Batchbald: efficient and diverse batch acquisition for deep Bayesian active learning (NeurIPS) (available at: https://papers.nips.cc/paper_files/paper/2019/hash/01e00f2bf7b066b285d343c2e8a6f5f1-Abstract.html)
Ma S, Wu H, Lawlor A and Dong R 2024 Breaking the barrier: selective uncertainty-based active learning for medical image segmentation (arXiv: 2401.16298)
Mehta R *et al* 2022 Qu-brats: MICCAI brats 2020 challenge on quantifying uncertainty in brain tumor segmentation – analysis of ranking scores and benchmarking results *J. Mach. Learn. Biomed. Imaging* **1** 1
Menze B H *et al* 2015 The multimodal brain tumor image segmentation benchmark (BRATS) *IEEE Trans. Med. Imaging* **34** 1993
Müller D, Soto-Rey I and Kramer F 2022 Towards a guideline for evaluation metrics in medical image segmentation *BMC Res. Notes* **15** 210
Ronneberger O, Fischer P and Brox T 2015 U-net: convolutional networks for biomedical image segmentation (MICCAI) (arXiv: 1505.04597)

Sandler M, Howard A, Zhu M, Zhmoginov A and Chen L-C 2018 Mobilenetv2: inverted residuals and linear bottlenecks *Proc. IEEE Conf. on Computer Vision and Pattern Recognition* pp 4510–20

Sener O and Savarese S 2018 Active learning for convolutional neural networks: a core-set approach (ICLR) (arXiv: 1708.00489)

Setiawan A W 2020 Image segmentation metrics in skin lesion: accuracy, sensitivity, specificity, dice coefficient, Jaccard index and Matthews correlation coefficient *2020 Int. Conf. on Computer Engineering, Network and Intelligent Multimedia (CENIM)* (IEEE) pp 97–102

Settles B 2009 Active learning literature survey *Technical Report* University of Wisconsin–Madison (available at: https://burrsettles.com/pub/settles.activelearning.pdf)

Simonyan K and Zisserman A 2014 Very deep convolutional networks for large-scale image recognition, (arXiv: 1409.1556)

Sinha S, Ebrahimi S and Darrell T 2019 Vaal: variational adversarial active learning (ICCV) (https://doi.org/10.1109/ICCV.2019.00546)

Tan M and Le Q 2019 Efficientnet: rethinking model scaling for convolutional neural networks *Int. Conf. on Machine Learning Organization PMLR* pp 6105–14

Wang G, Li W, Aertsen M, Deprest J, Ourselin S and Vercauteren T 2019 Aleatoric uncertainty estimation with test-time augmentation for medical image segmentation with convolutional neural networks *Neurocomputing* **338** 34

Wu X, Gao Y, Ma Y, Zhang J and Liu Y 2024 Multi-faceted consistency learning with active cross-annotation for barely supervised 3D brain tumor segmentation MICCAI (available at: https://papers.miccai.org/miccai-2024/paper/0102_paper.pdf)

Yoo D and Kweon I S 2019 Learning loss for active learning (CVPR) (available at: https://openaccess.thecvf.com/content_CVPR_2019/html/Yoo_Learning_Loss_for_Active_Learning_CVPR_2019_paper.html)

Zhou T *et al* 2024 SBC-AL: structure and boundary consistency-based active learning for medical image segmentation (MICCAI) (available at: https://papers.miccai.org/miccai-2024/paper/3047_paper.pdf)